



life.augmented

Debug Authentication

Tomas Team

- Timing : 1 hour
- Purpose
 - Clarify this new feature with details
 - Show how DA can help customers in field return analysis
- Process
 - First glance: reopen debugging link as if it was a field return
 - Explain in detail the Debug Authentication feature
 - Practice with a typical use case
- Payoff
 - Understand the Debug authentication mechanism
 - Understand that SM benefits from the platform security mechanisms



Introduction

- Our product is on the field, unfortunately some devices are not working fine...
- Customer returned the product to the support team
- Let's see how the support team can reopen the device.

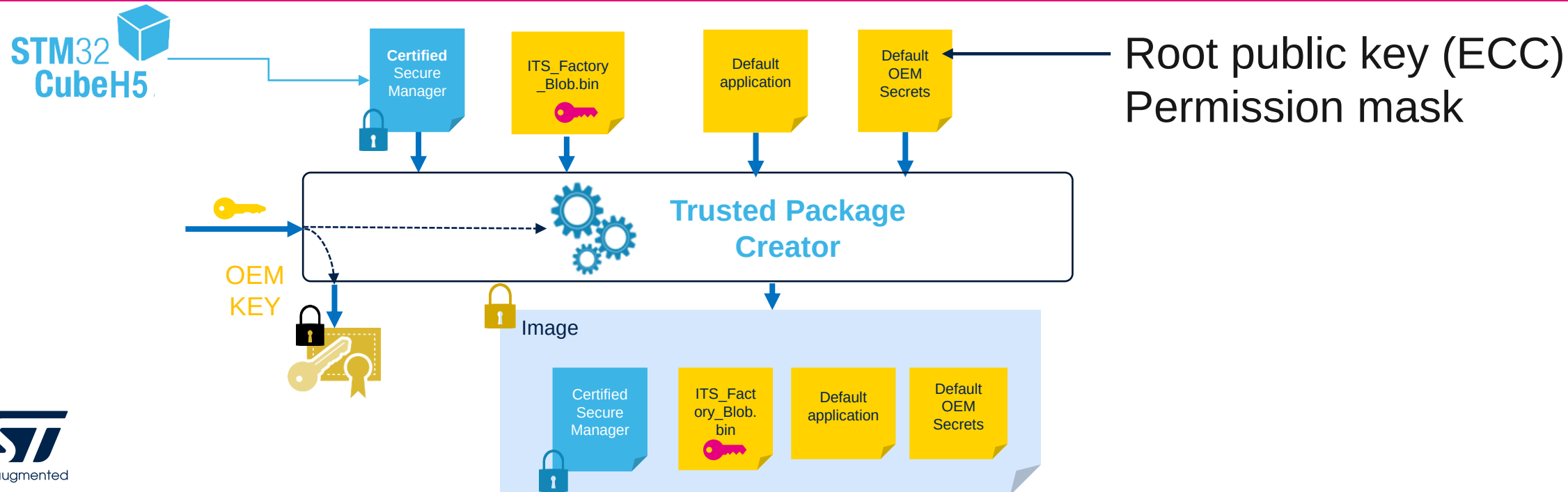


What has been provisioned in the product?

* DA configuration:

```
Warning: Default keys must NOT be used in a product. Make sure to regenerate your own keys
From TrustedPackageCreator (tab H5-DA CertifGen),
update the keys(s) (in \ROT_Provisioning\DA\Keys) and permissions (if/as needed)
then regenerate the certificate(s)
From TrustedPackageCreator (tab H5-OBKey),
Select DA_Config.xml (in \ROT_Provisioning\DA\Config)
Update the configuration (if/as needed) then generate DA_Config.obk file
Press any key to continue...
```

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Keys\key_1_root_pub.pem





First possibility: re-open the debug

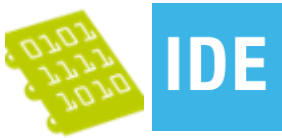
- What do we need ?
 - A certificate that allows reopening the device for debug and regression (leaf certificate)

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Certificates\cert_leaf_chain.b64

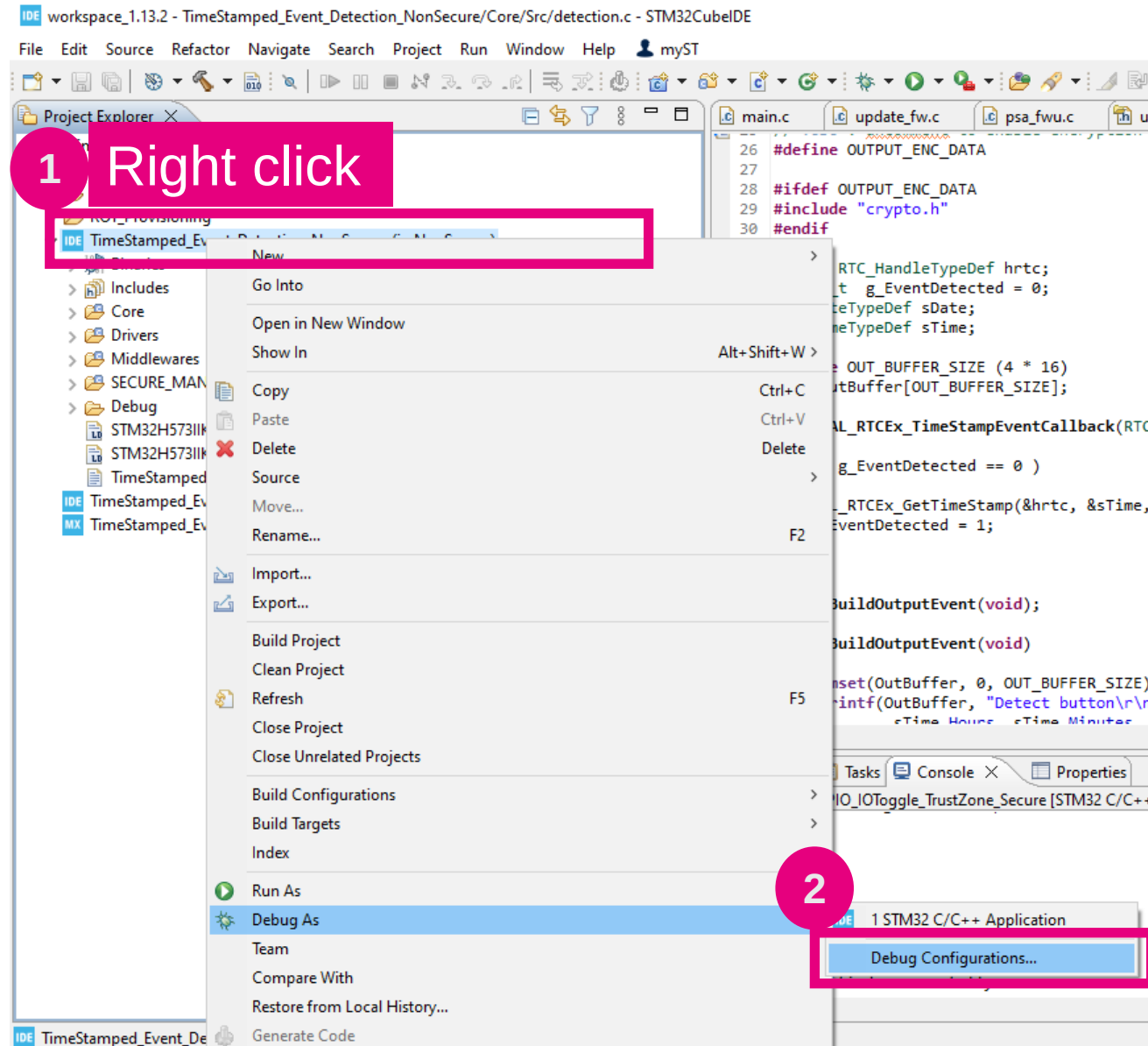
- The associated private ECC key

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Keys\key_3_leaf.pem

Let's reopen the device for debug !



Modify the debug configuration



We will connect in hotplug...

IDE Debug Configurations

Create, manage, and run configurations

type filter text

- C++ Application
- C++ Attach to Application
- C++ Postmortem Debugger
- C++ Remote Application
- IB Hardware Debugging
- unch Group
- M32 C/C++ Application
- TimeStamped_Event_Detection_NonSecure

1

Name: Stamped_Event_Detection_NonSecure

Main **Debugger** Startup Source Common

GDB Connection Settings

☒ Autostart local GDB server Host name or IP address: localhost

☐ Connect to remote GDB server Port number: 61234

Debug probe: ST-LINK (ST-LINK GDB server)

GDB Server Command Line Options

Show Command Line

Interface

☒ SWD ☐ JTAG

☐ ST-LINK S/N: Scan

Frequency (kHz): Auto

Access port: 1 - Cortex-M33

2

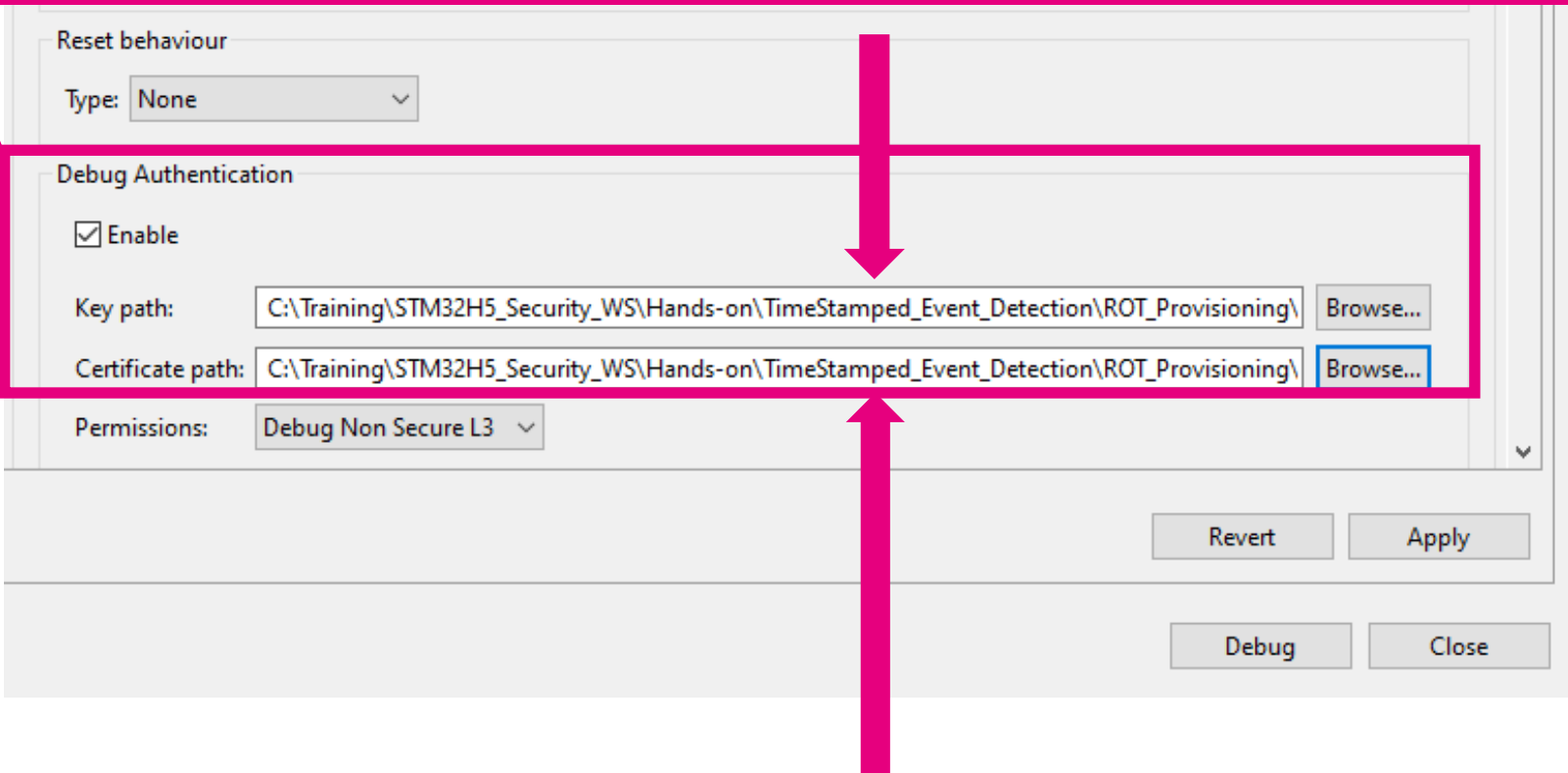
Reset behaviour

Type: None

Give input for Debug authentication

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Keys\key_3_leaf.pem

3



Reset behaviour
Type: None

Debug Authentication

☒ Enable

Key path: C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\ Browse...

Certificate path: C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\ Browse...

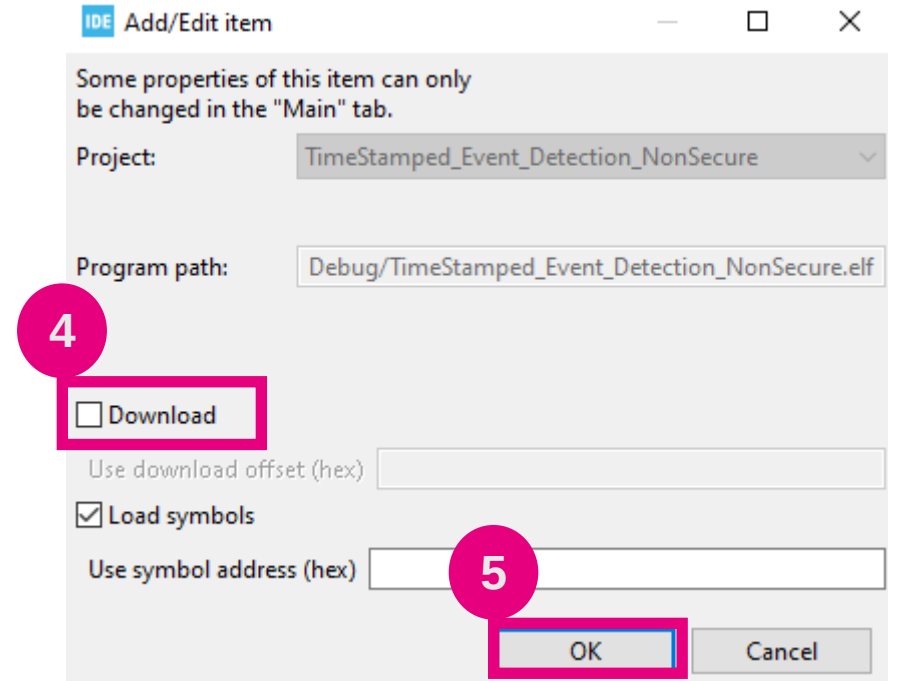
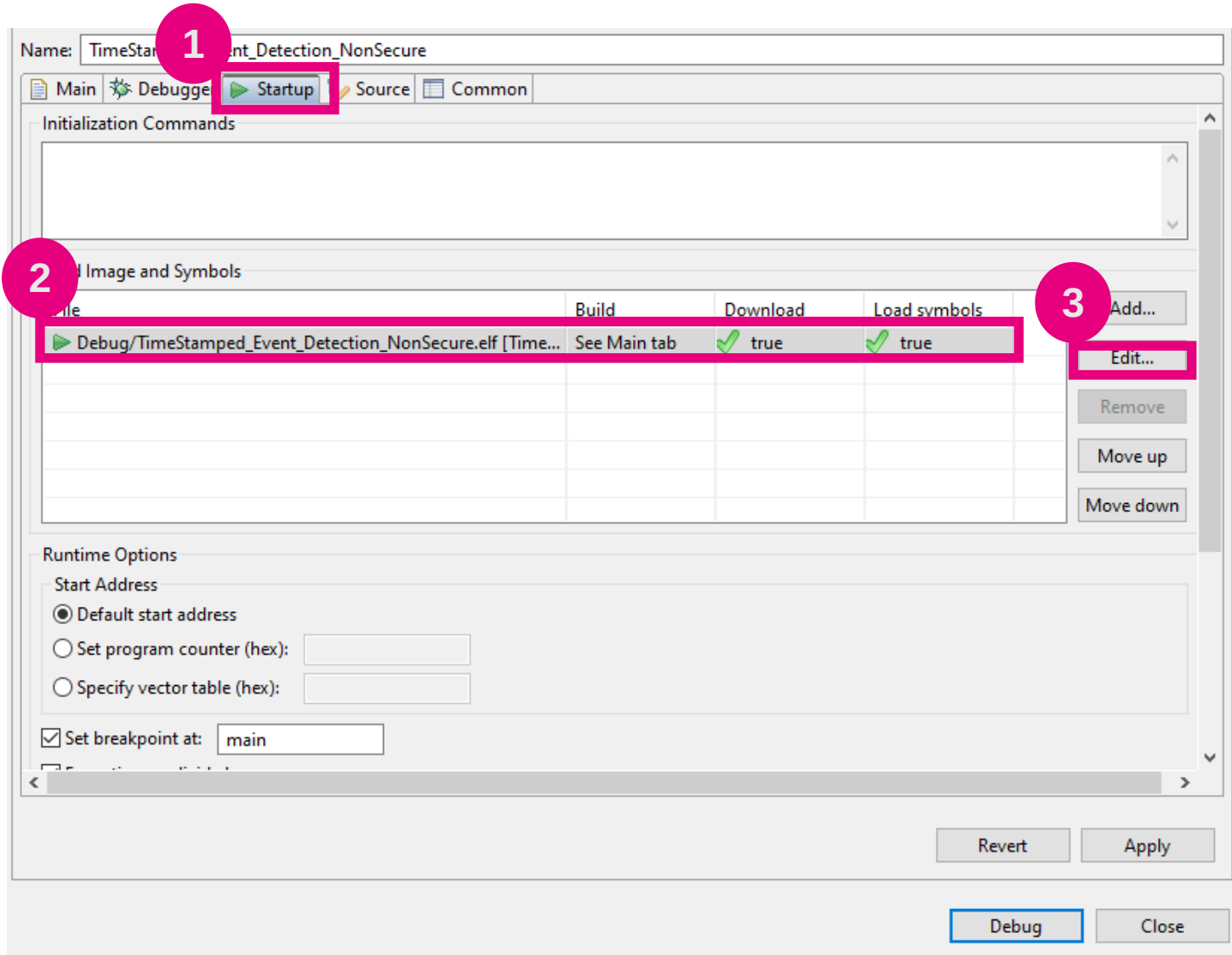
Permissions: Debug Non Secure L3

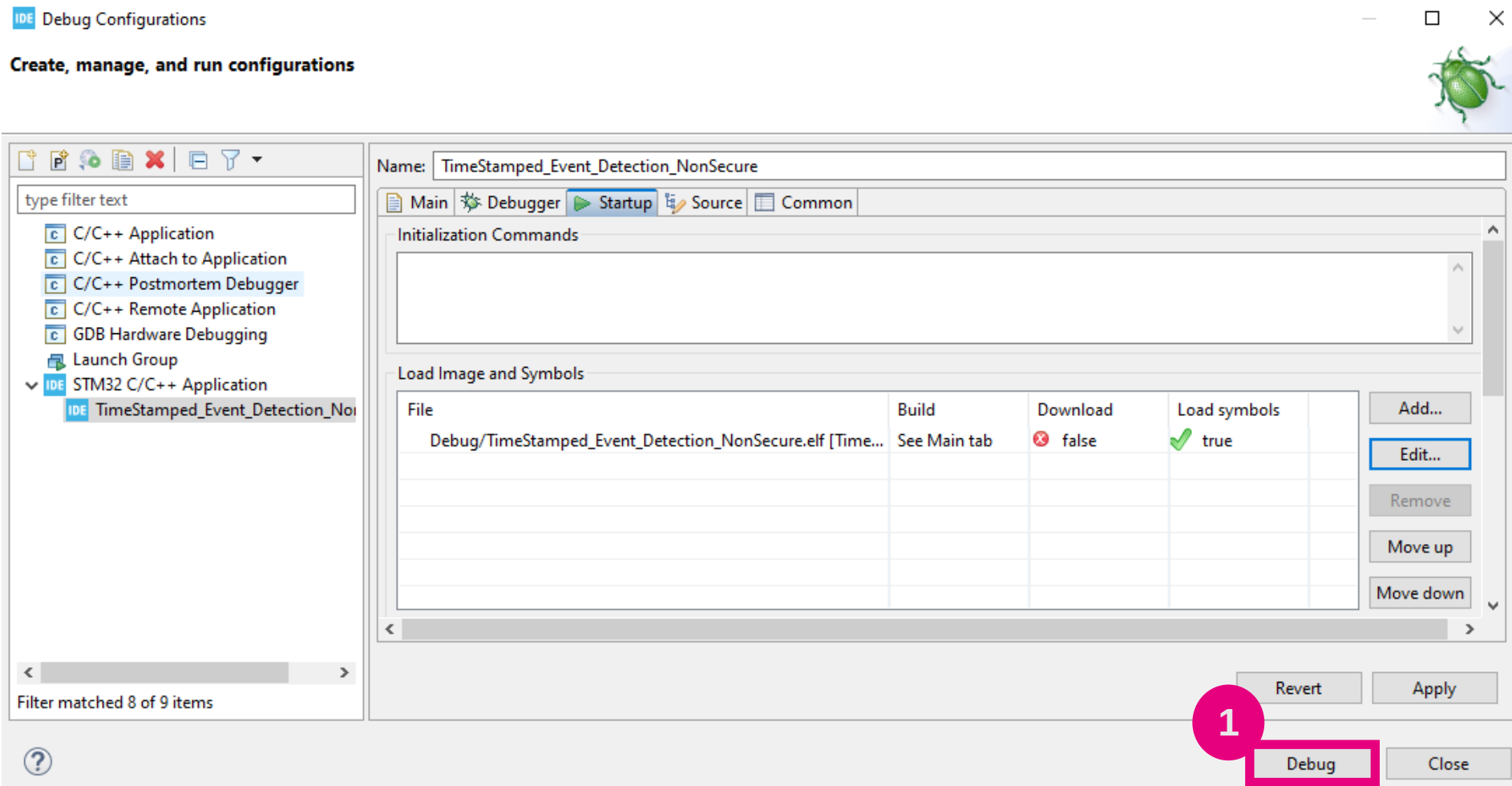
Revert Apply

Debug Close

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Certificates\cert_leaf_chain.b64

Remove the download of the firmware





Break execution

workspace_1.13.2 - TimeStamped_Event_Detection_NonSecure/Drivers/STM32H5xx_HAL_Driver/stm32h5xx_hal_uart.c - STM32CubeIDE

File Edit Source Refactor N Search Project Run Window Help myST

Debug Project Explorer

TimeStamped_Event_Detection_NonSecure [STM32 C/C++ Appli]
 TimeStamped_Event_Detection_NonSecure.elf [cores: 1]
 Thread #1 [main] 1 [core: 1] (Running : User Request)
 arm-none-eabi-gdb (12.1.90.20220802)
 ST-LINK (ST-LINK GDB server)

```

3455 if (__HAL_UART_GET_FLAG(huart, UART_FLAG_RTOF) == SET)
3456 {
3457     /* Clear Receiver Timeout flag*/
3458     __HAL_UART_CLEAR_FLAG(huart, UART_CLEAR_RTOF);
3459
3460     /* Blocking error : transfer is aborted
3461     Set the UART state ready to be able to start again the process,
3462     Disable Rx Interrupts if ongoing */
3463     UART_EndRxTransfer(huart);
3464
3465     huart->ErrorCode = HAL_UART_ERROR_RTO;
3466
3467     /* Process Unlocked */
3468     __HAL_UNLOCK(huart);
3469
3470     return HAL_TIMEOUT;
3471 }
3472 }
3473 }
3474 }
3475 return HAL_OK;
3476 }
3477
3478 /**
3479  * @brief Start Receive operation in interrupt mode.
3480  * @note This function could be called by all HAL UART API providing reception in Interrupt
3481  * @note When calling this function, parameters validity is considered as already checked,
3482  * i.e. Rx State, buffer address, ...
3483  * UART Handle is assumed as locked
  
```

Console Problems Executables Debugger Console Memory

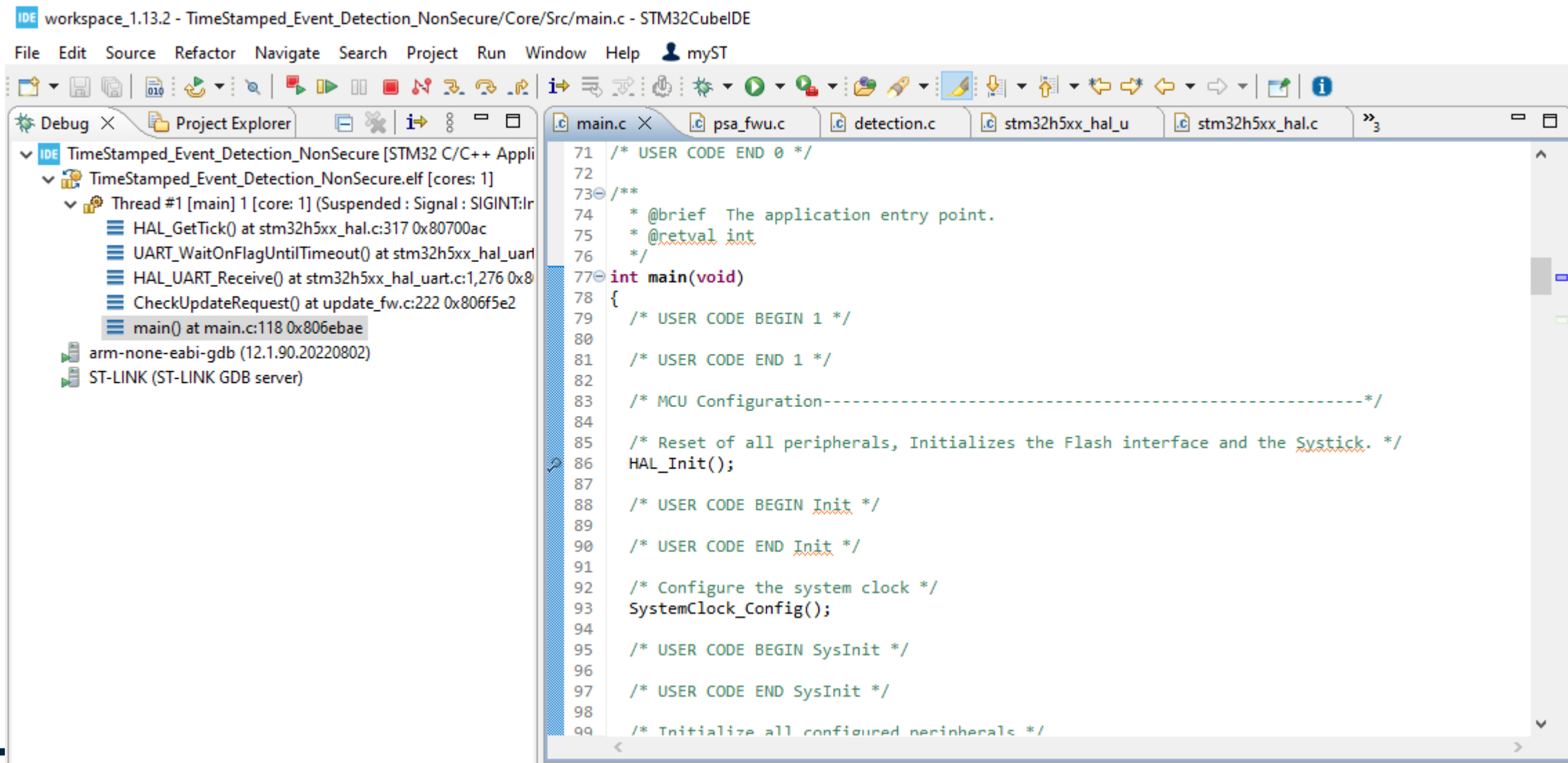
TimeStamped_Event_Detection_NonSecure [STM32 C/C++ Application] [pid: 106]

```

Listen Port Number      : 61254
Status Refresh Delay    : 15s
Verbose Mode            : Disabled
SWD Debug               : Enabled

Waiting for debugger connection...
Debugger connected
Waiting for debugger connection...
Debugger connected
Waiting for debugger connection...
  
```

You can set breakpoint and debug...





Some explanation

- We have seen that we can reopen the debug on a closed device
- This can be done thanks to Debug Authentication mechanism
- Let's see in more details how this works



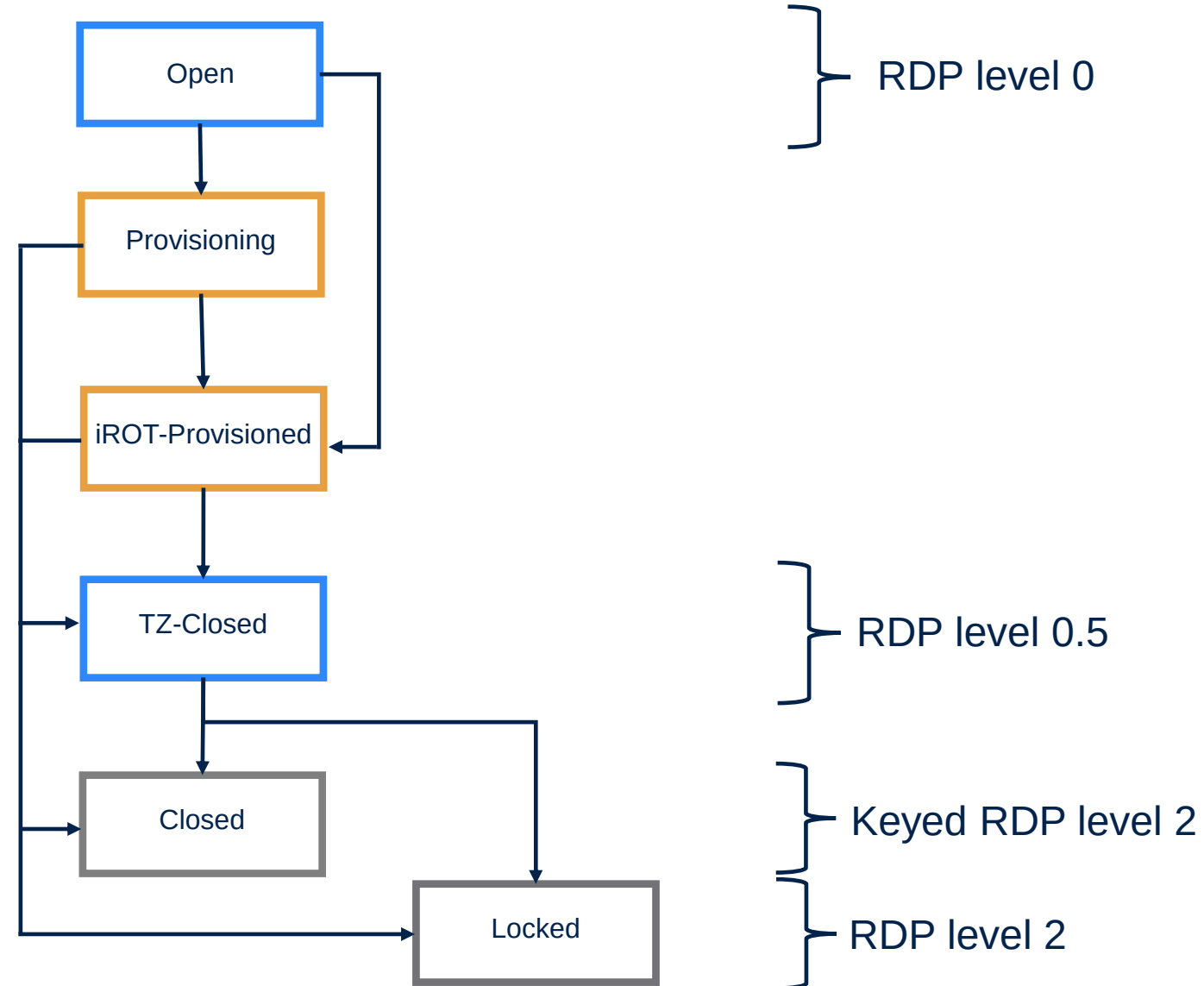
STM32H5 Product Life Cycle Quick overview

- Product state : First introduced at beginning of the workshop

Open	State fully open (default)
Provisioning	State to provision OB keys
iROT-Provisioned	iRot has been provisioned
TZ-Closed	Only non secure is open. State only available if TZ is enabled
Closed	Final state but with controlled debug /regression capabilities
Locked	Final state without any reconnection possibility



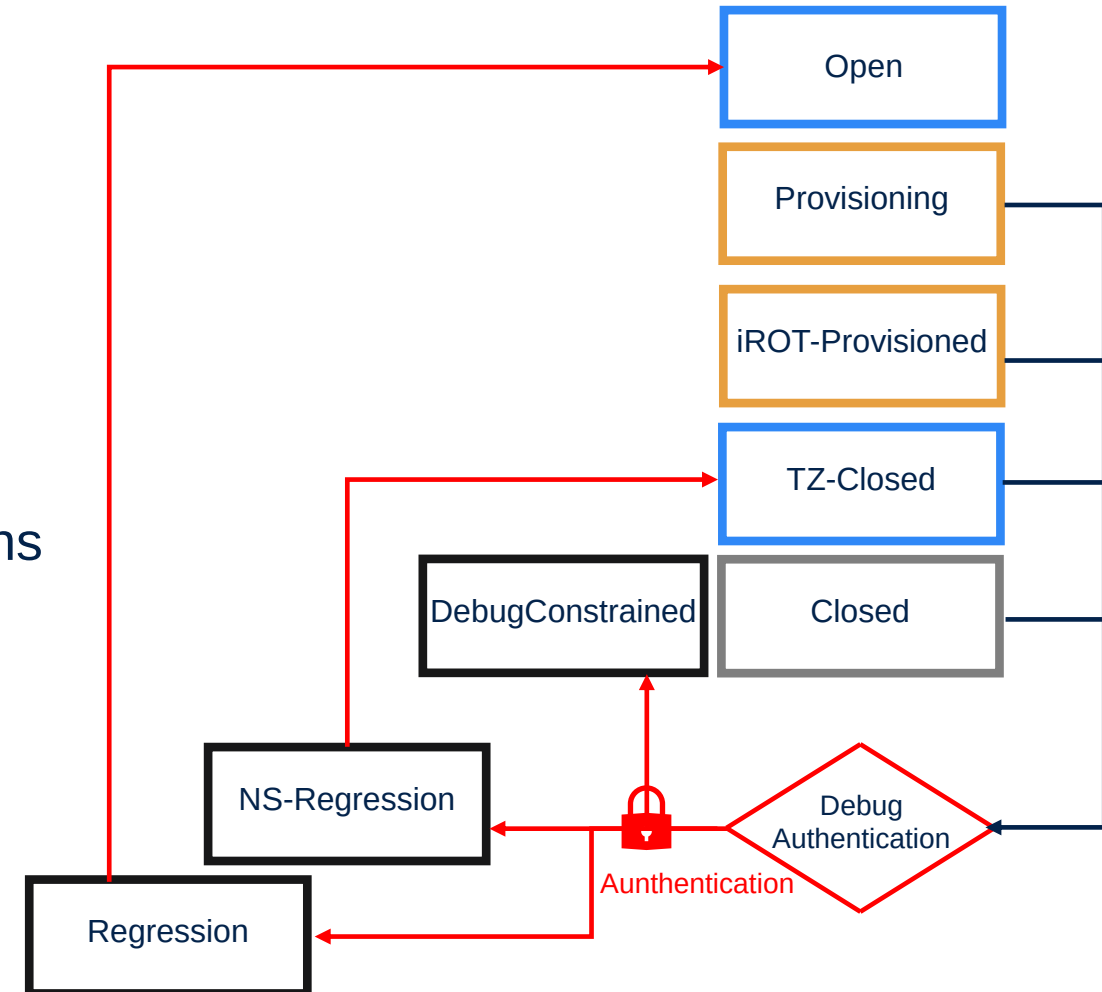
STM32H5 Product Life Cycle quick overview





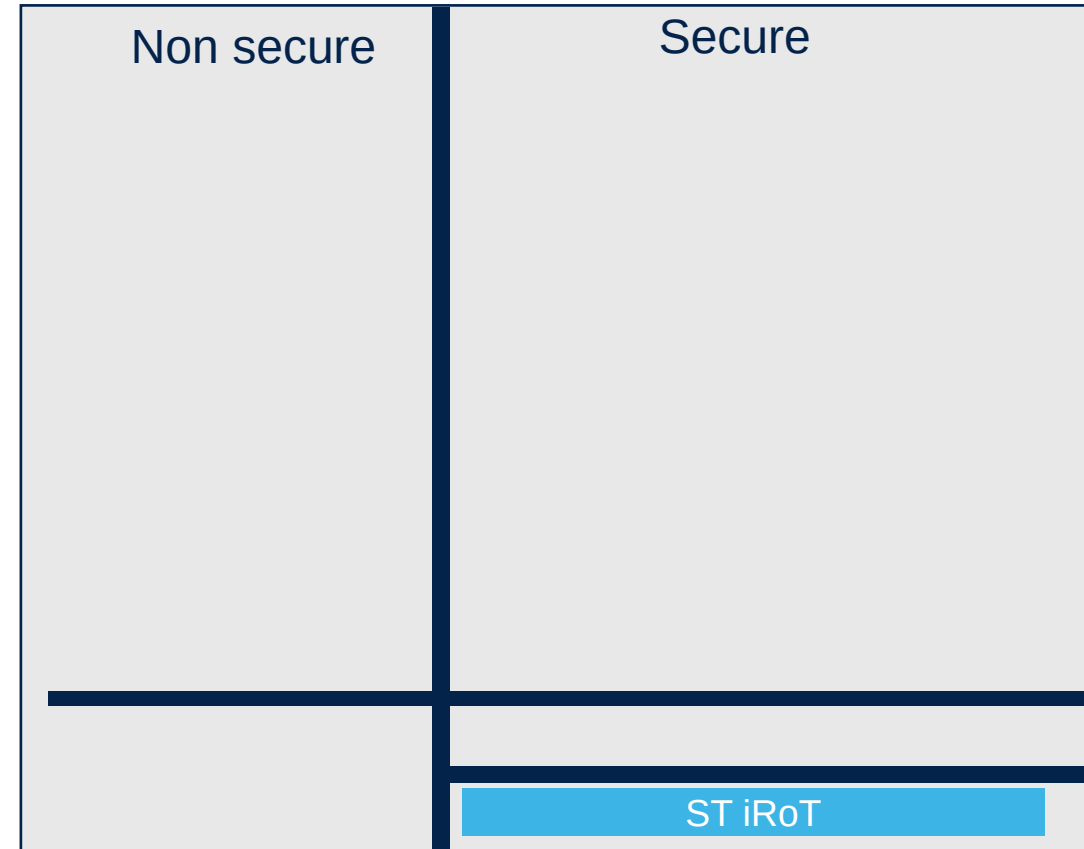
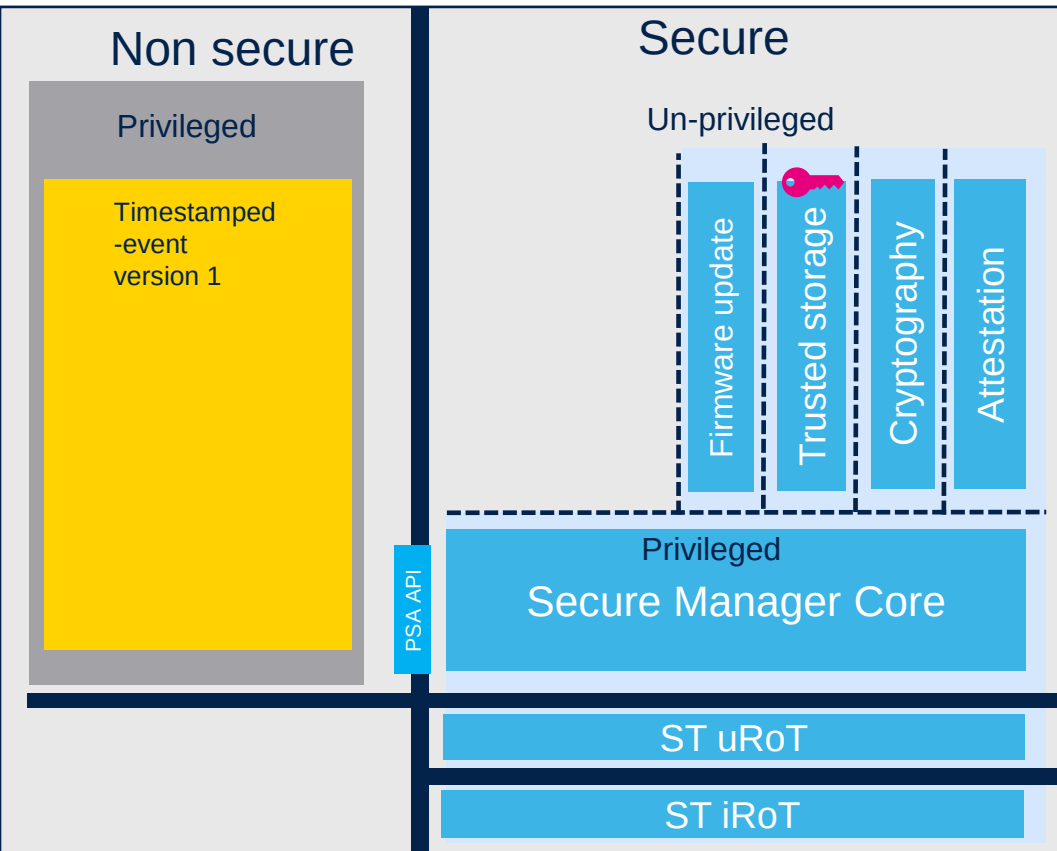
STM32H5 Security Debug Authentication Control

- Debug Authentication allows to control
 - Re-enabling debug possibility
 - Partial or full regression
- Need to provision STM32 with :
 - ECC pub key to allow this debug authentication
 - SOC_PERMISSION: 16 bits defining the permissions authorized by default
- Need to create a certificate, to be able to authenticate



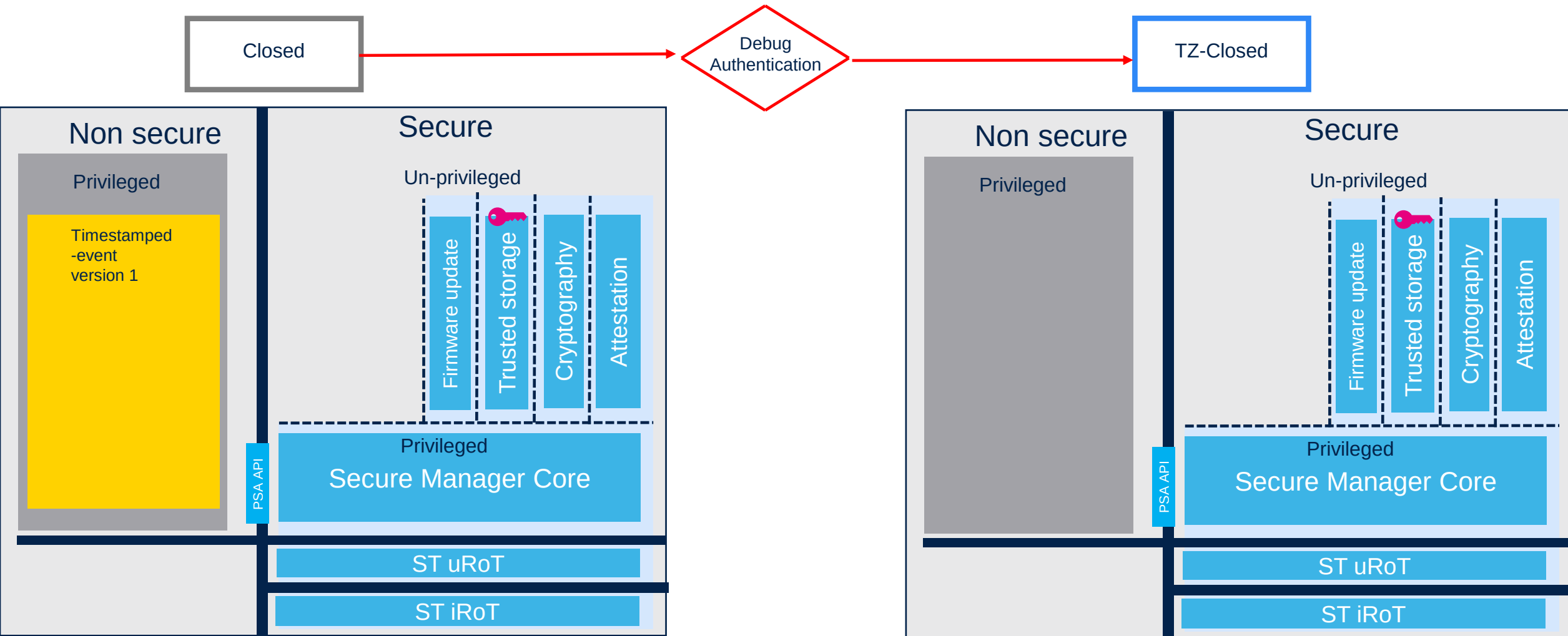


STM32H5 Regression





STM32H5 NS-Regression (partial)





STM32H5 Security Debug Authentication Provisioning



SOC Permission Mask: bitmask which defines what could be done thanks to DA access on this target

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
X		X		X	X	X	X	X				X			

Regressions (full or partial)

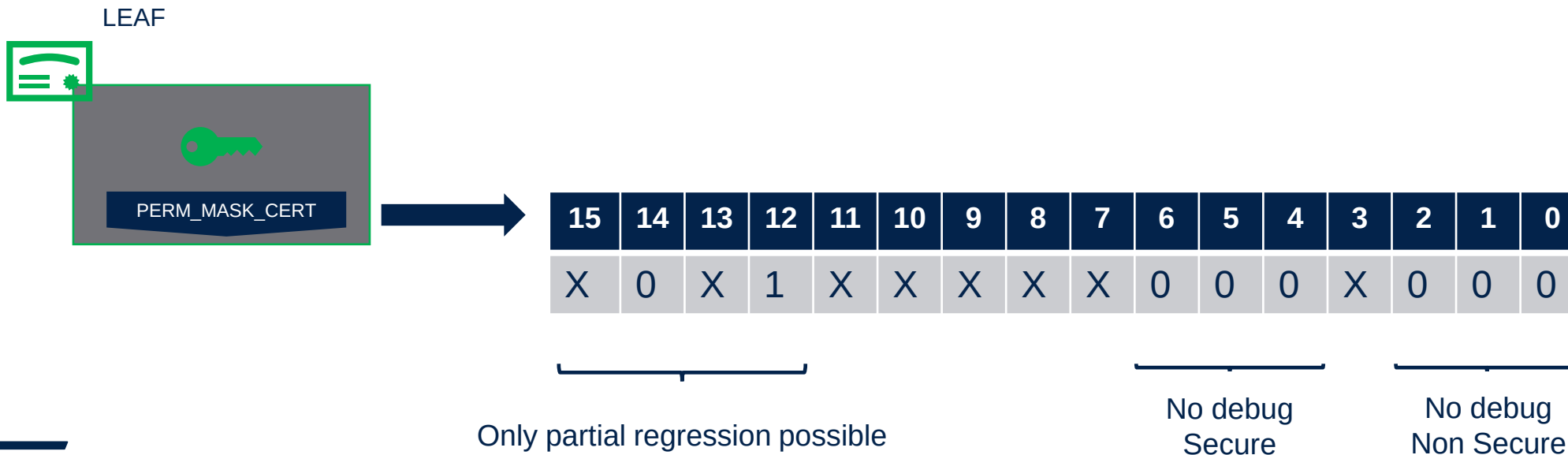
Intrusive
Secure mode

Intrusive
Non Secure



What is needed to do a Debug Authentication?

- A private ECC key 
- A certificate LEAF that embeds :
 - An ECC public key 
 - A permission mask which defines the capability associated to the certificate
 - For example : this certificate only allows partial regression

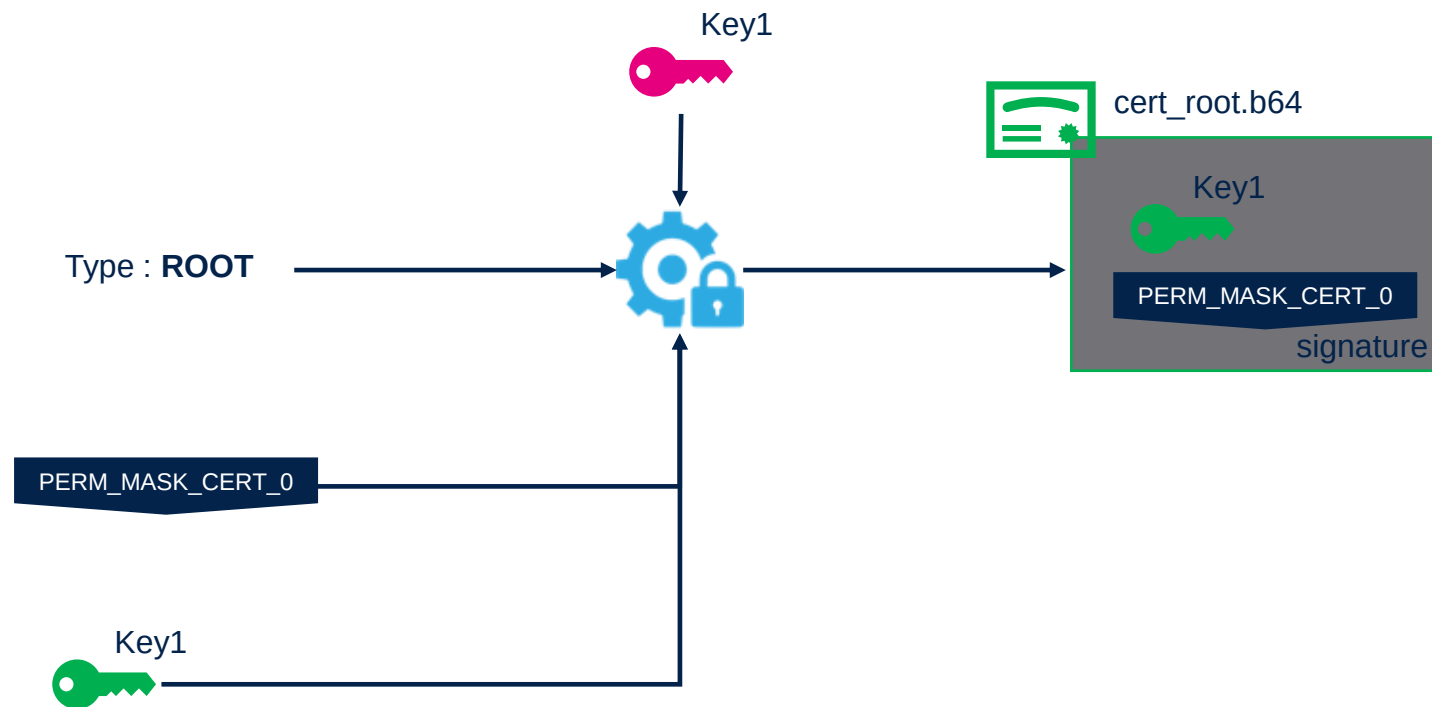




How to generate a certificate to do a debug authentication ?

Reminder :

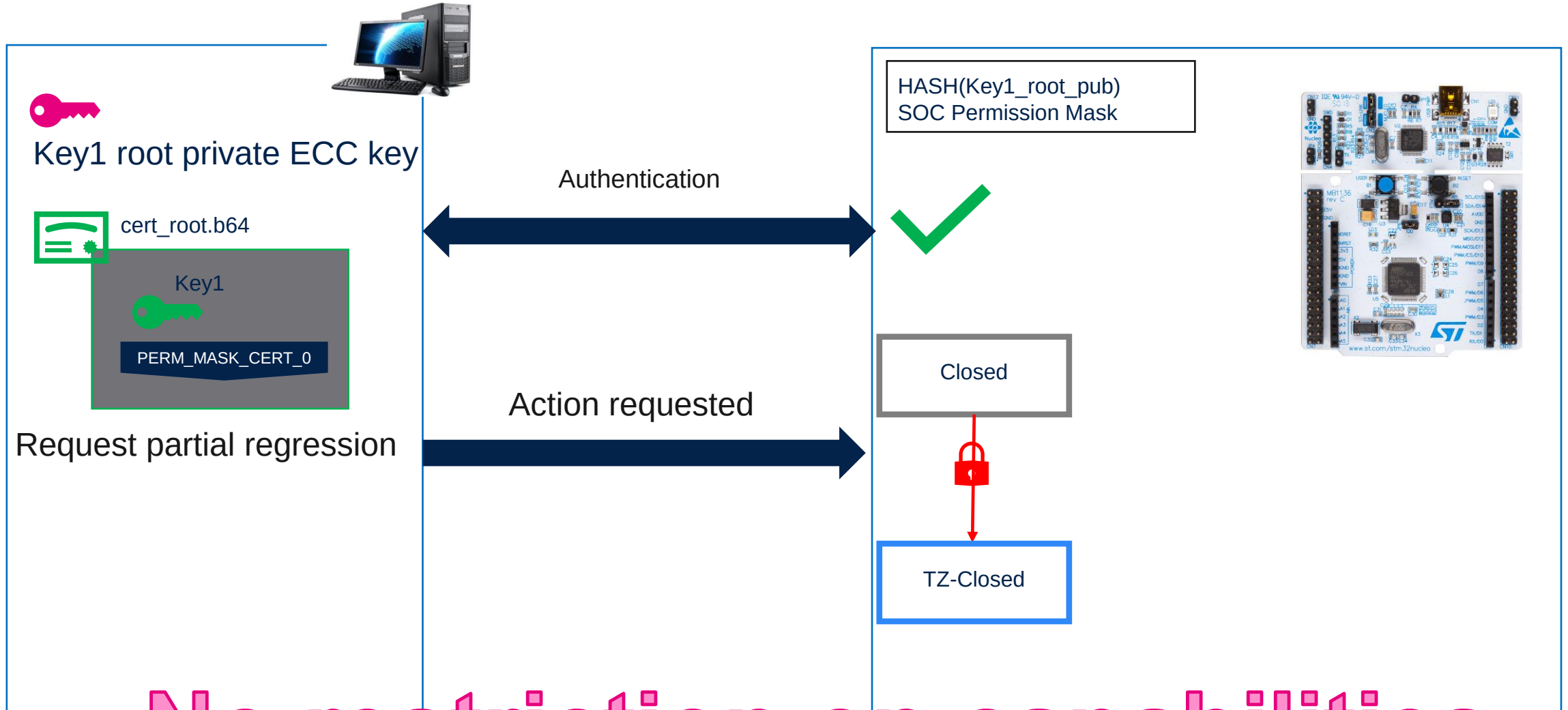
- Private key is used to generate signature
- Public key is used to check signature
- Check the signature of the certificate guarantee the authenticity and the integrity of this one.





STM32H5 Security

Debug Authentication partial regression

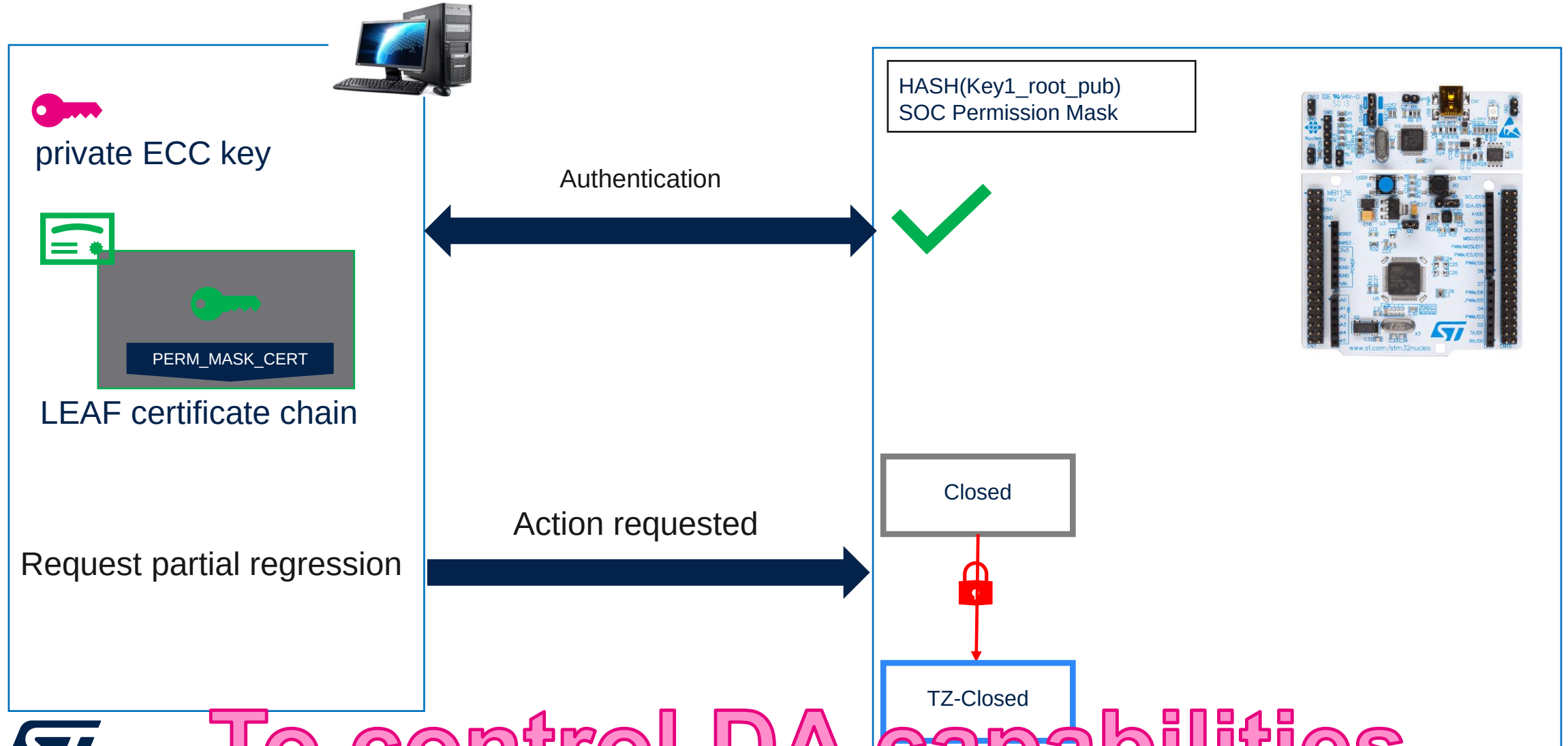


No restriction on capabilities...



STM32H5 Security

Debug Authentication partial regression

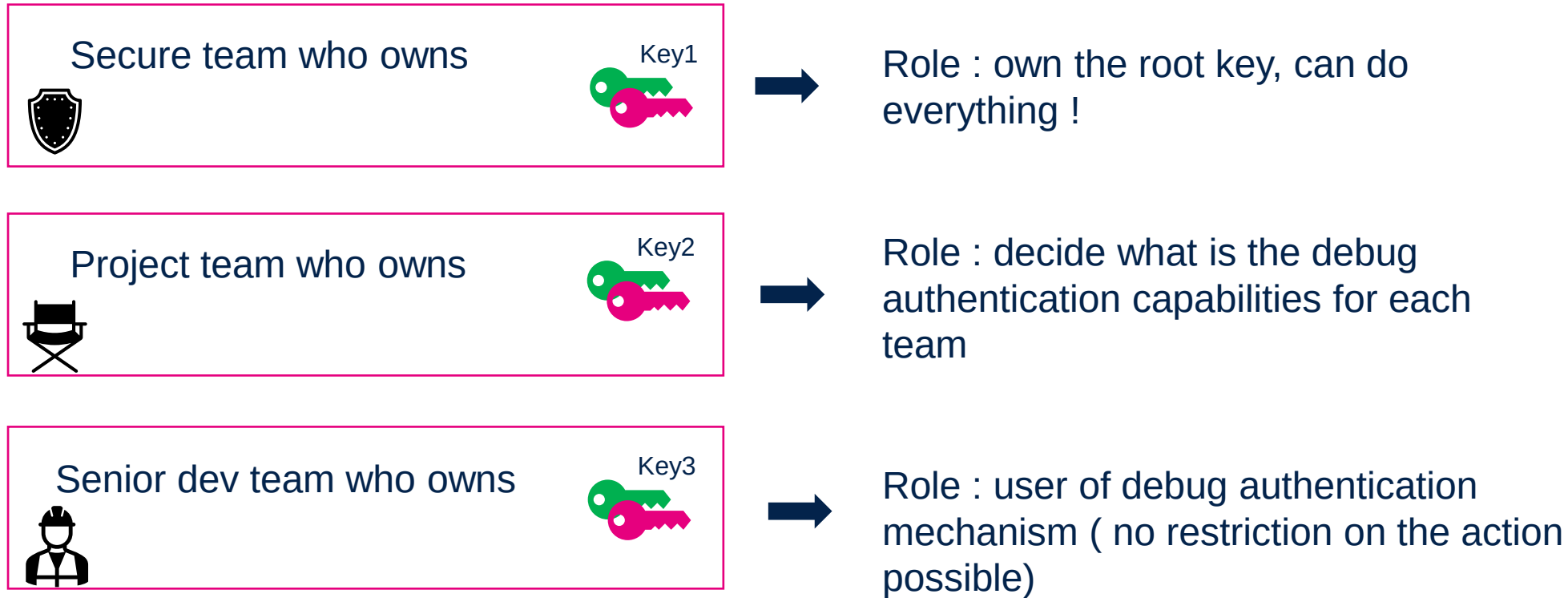


To control DA capabilities...



STM32H5 Security

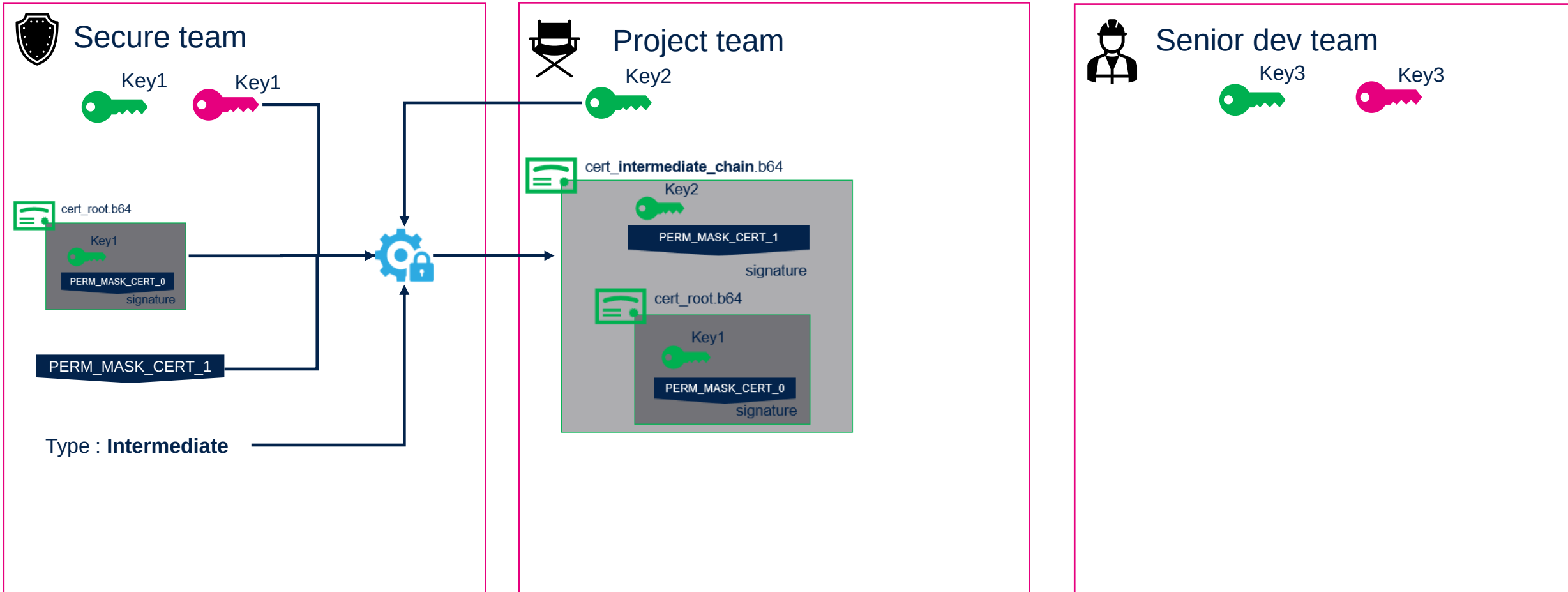
Leaf certificate embedded the certificate chain





STM32H5 Security

Create an intermediate certificate chain

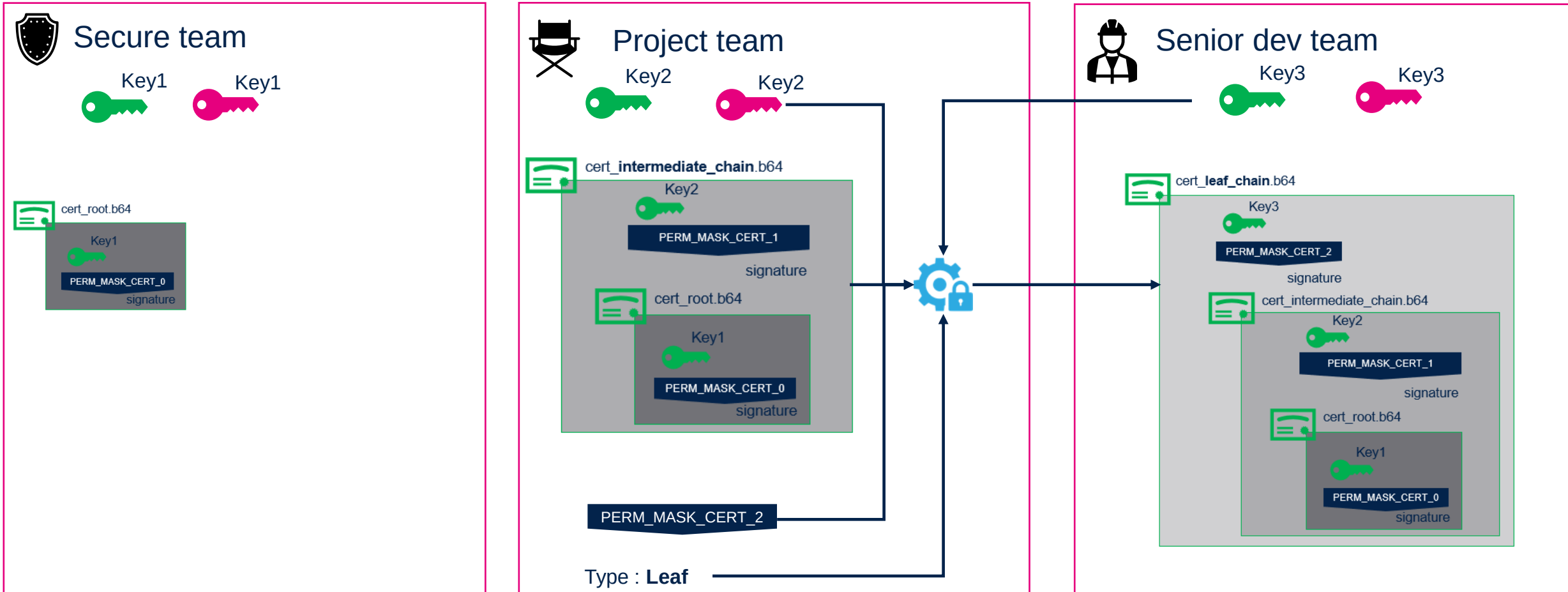


Why doing this action ?
Give the capability of Project team to
create certificate for other teams



STM32H5 Security

Create a leaf certificate chain



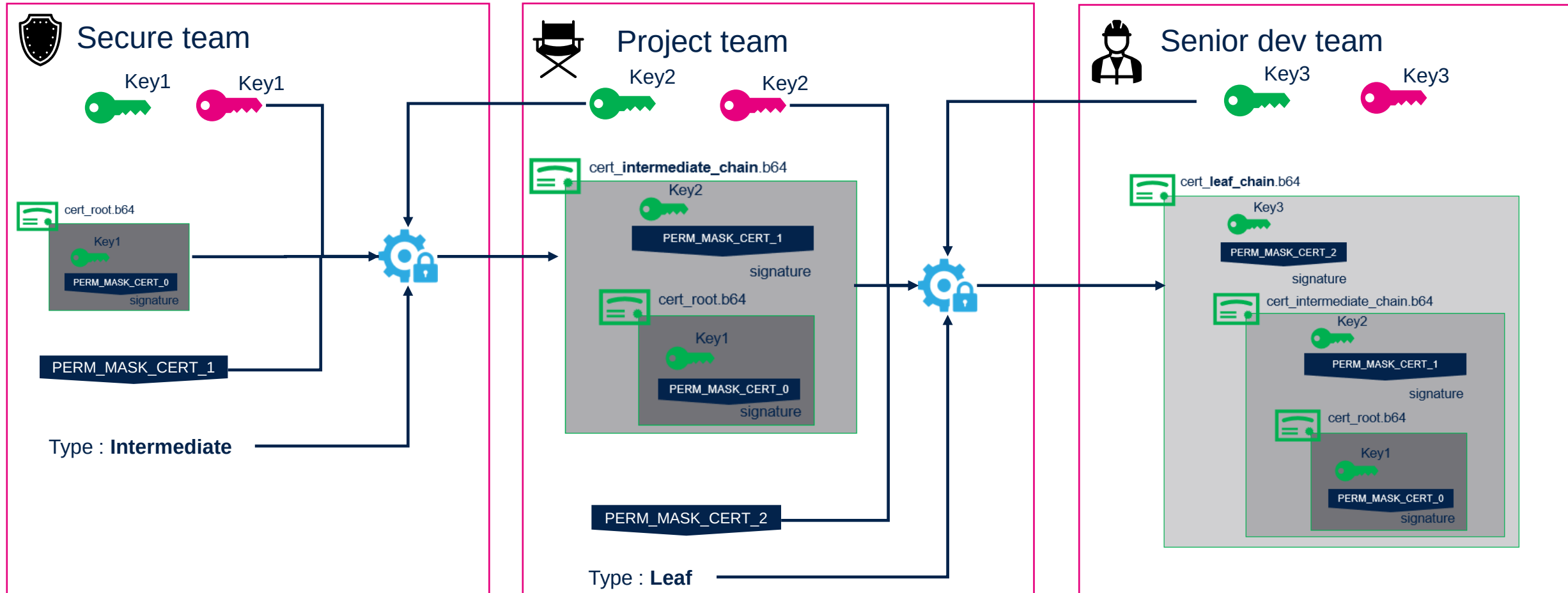
Why doing this action ?

Create a certificate chain that could be used by dev team with its private key



STM32H5 Security

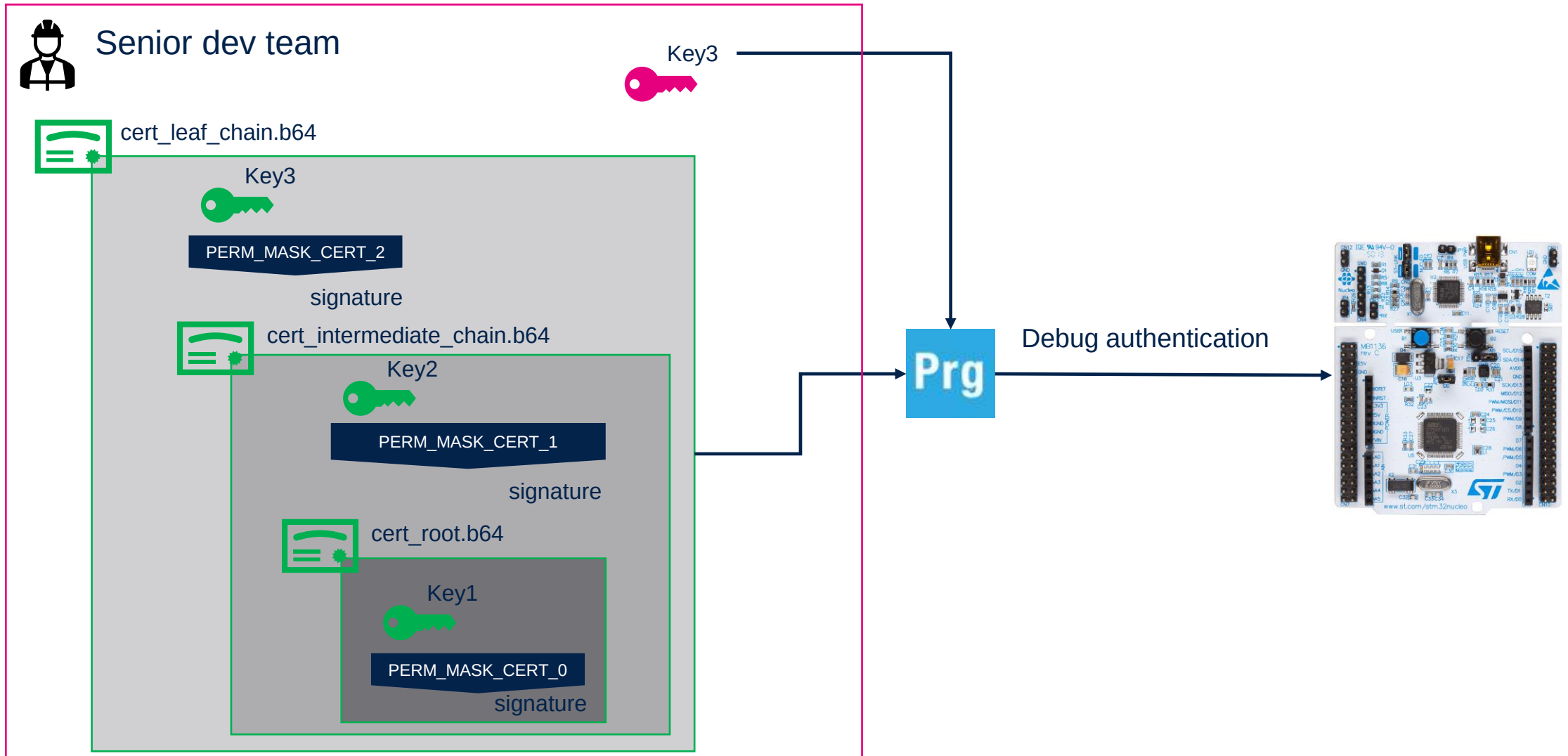
Certificate chain creation





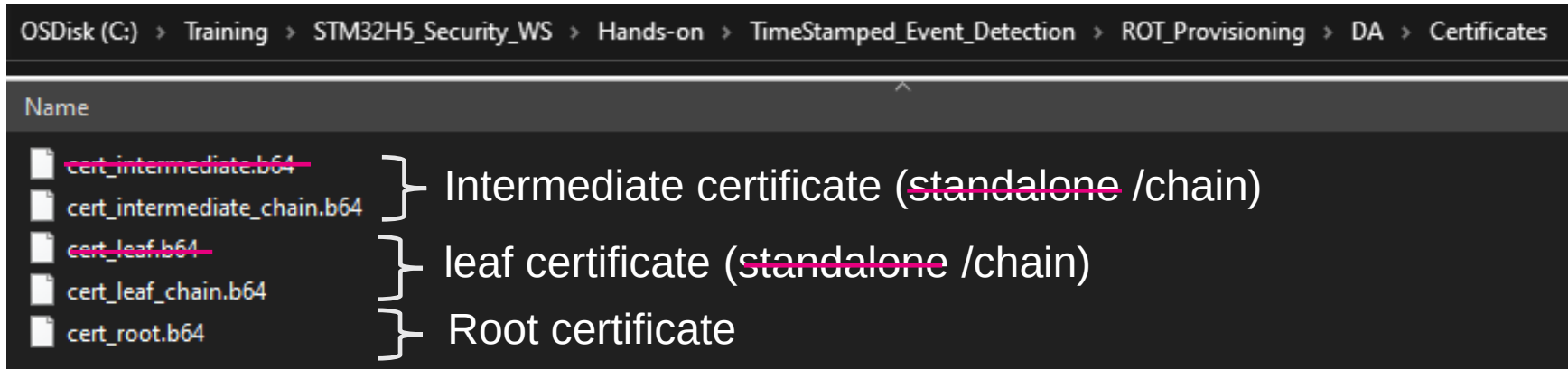
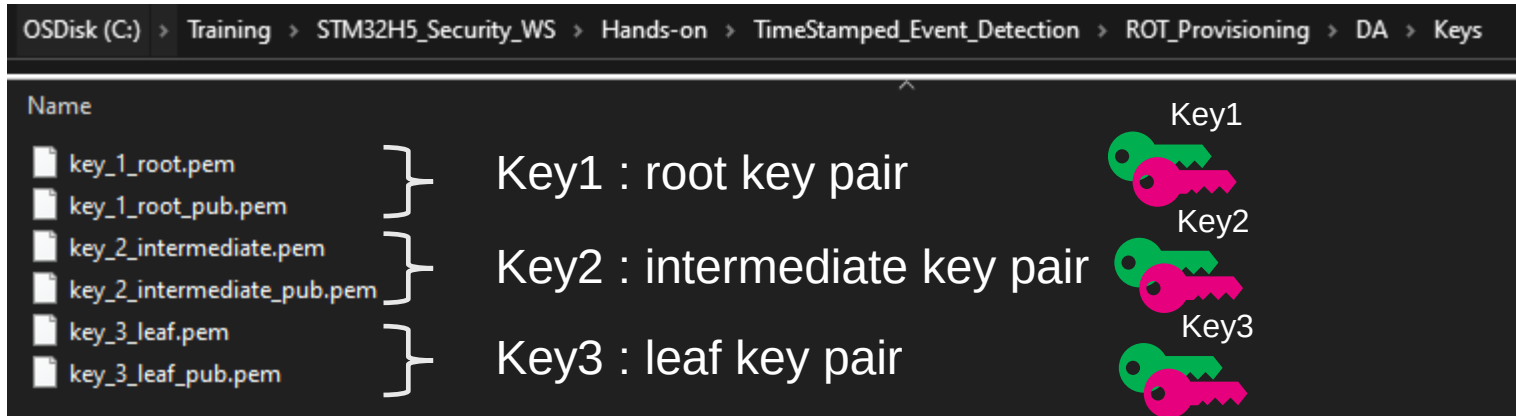
STM32H5 Security

Debug authentication with certificate leaf chain





What is available in our project regarding DA ?



Type of certificate:

Root : used to create another certificate intermediate or leaf

Intermediate chain : used to create another certificate chain (standalone not used)

Leaf chain: can only be used for Debug Authentication (standalone not used)



Permission mask chaining

OSDisk (C:) > Training > STM32H5_Security_WS > Hands-on > TimeStamped_Event_Detection > ROT_Provisioning > DA > Certificates

Name

cert_intermediate.b64	}	Intermediate certificate
cert_intermediate_chain.b64		
cert_leaf.b64	}	leaf certificate
cert_leaf_chain.b64		
cert_root.b64	}	Root certificate

Type of certificate:

Root : use to create other certificate

PERM_MASK_CERT_0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
X	1	X	1	X	X	X	X	X	1	1	1	X	1	1	1

Intermediate : use to create other certificates

PERM_MASK_CERT_INTER_1

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
X	1	X	1	X	X	X	X	X	1	1	1	X	1	1	1

Leaf : can only be used to for Debug Authentication

PERM_MASK_CERT_2

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
X	1	X	1	X	X	X	X	X	1	1	1	X	1	1	1



15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
X	1	X	1	X	X	X	X	X	1	1	1	X	1	1	1

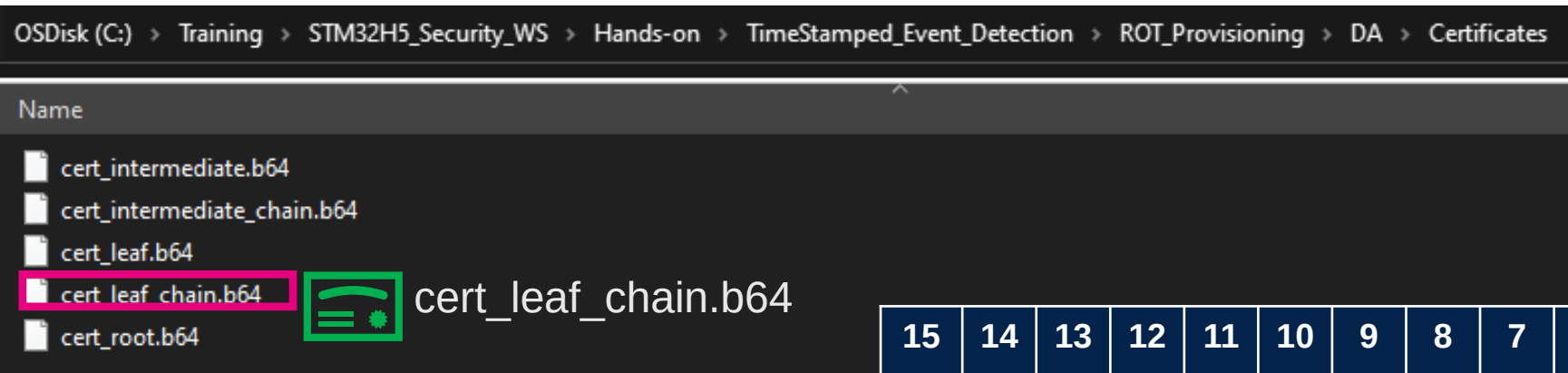
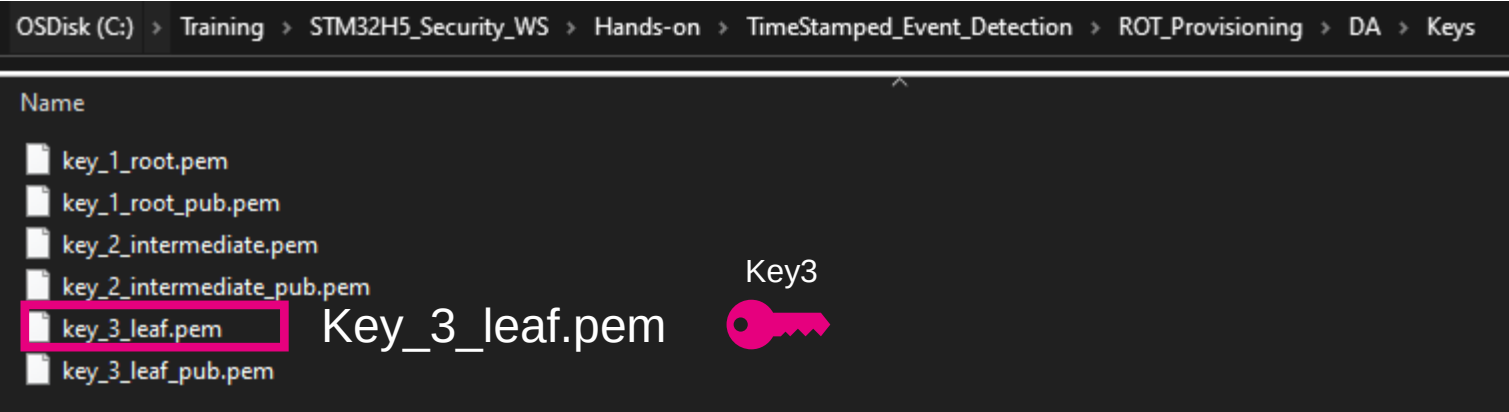
All regression possible

debug
Secure

debug
Non Secure



What can we do ?



ALL action are possible :

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
X	1	X	1	X	X	X	X	X	1	1	1	X	1	1	1

regression possible

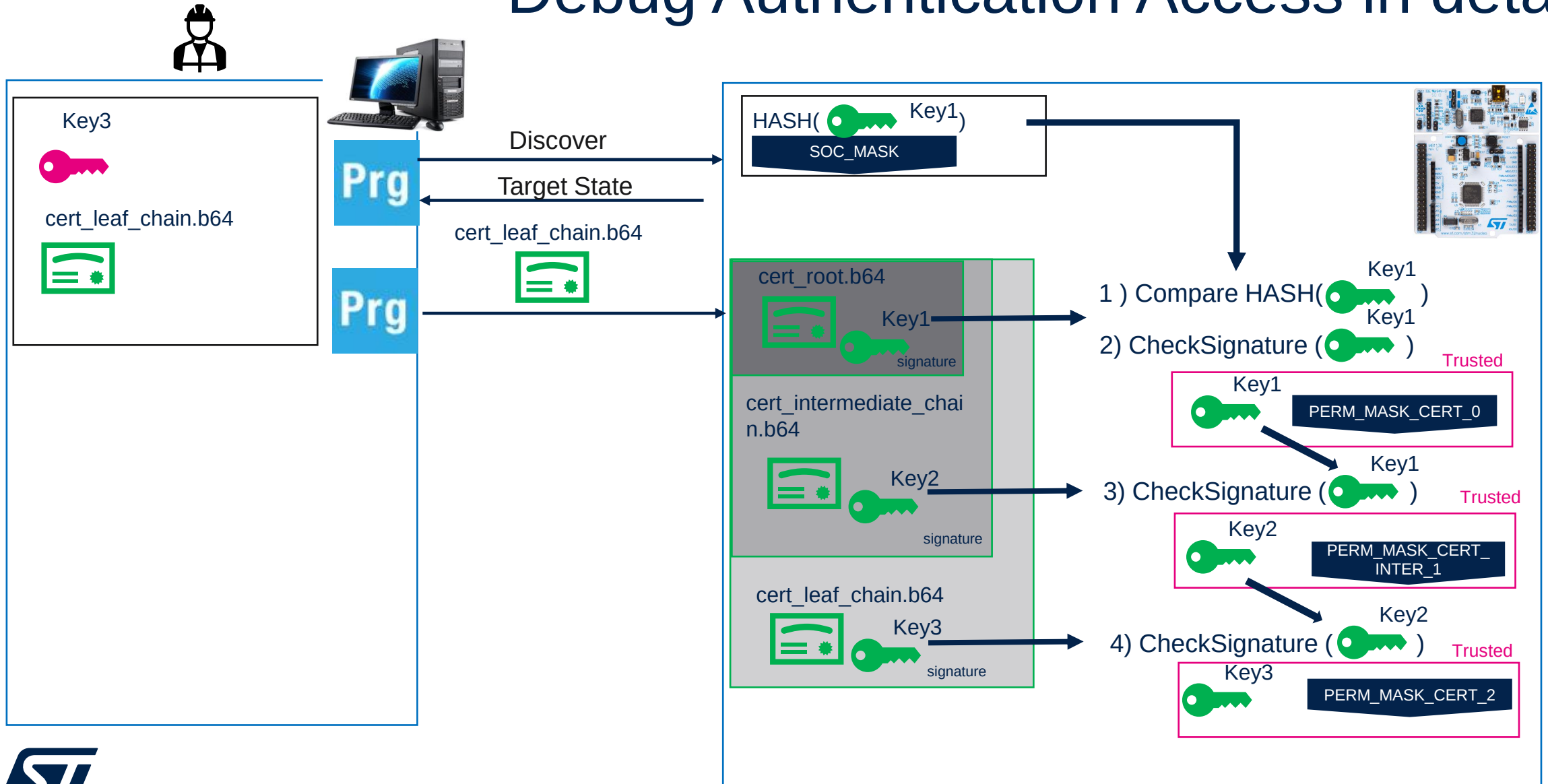
debug
Secure

debug
Non Secure



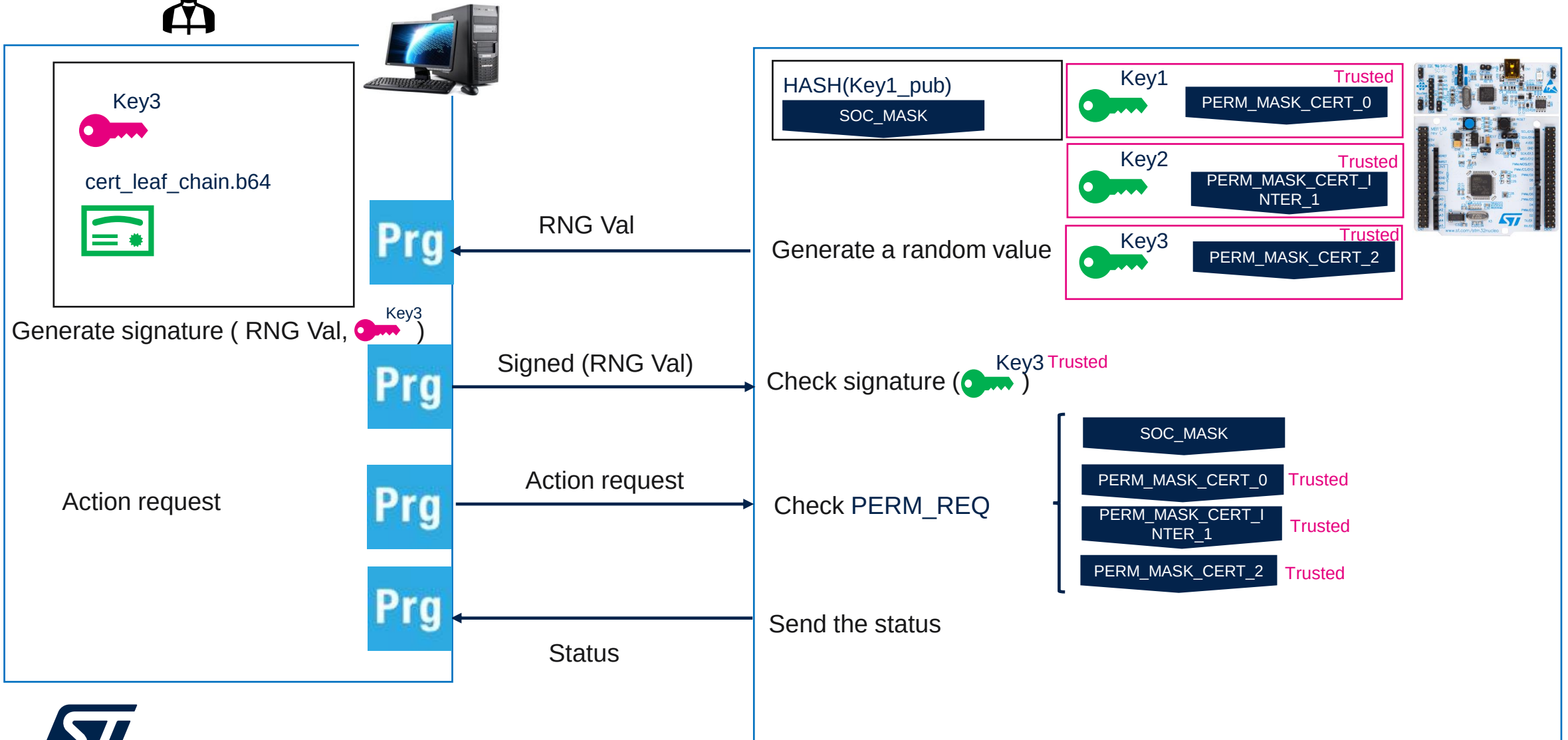
STM32H5 Security

Debug Authentication Access in details





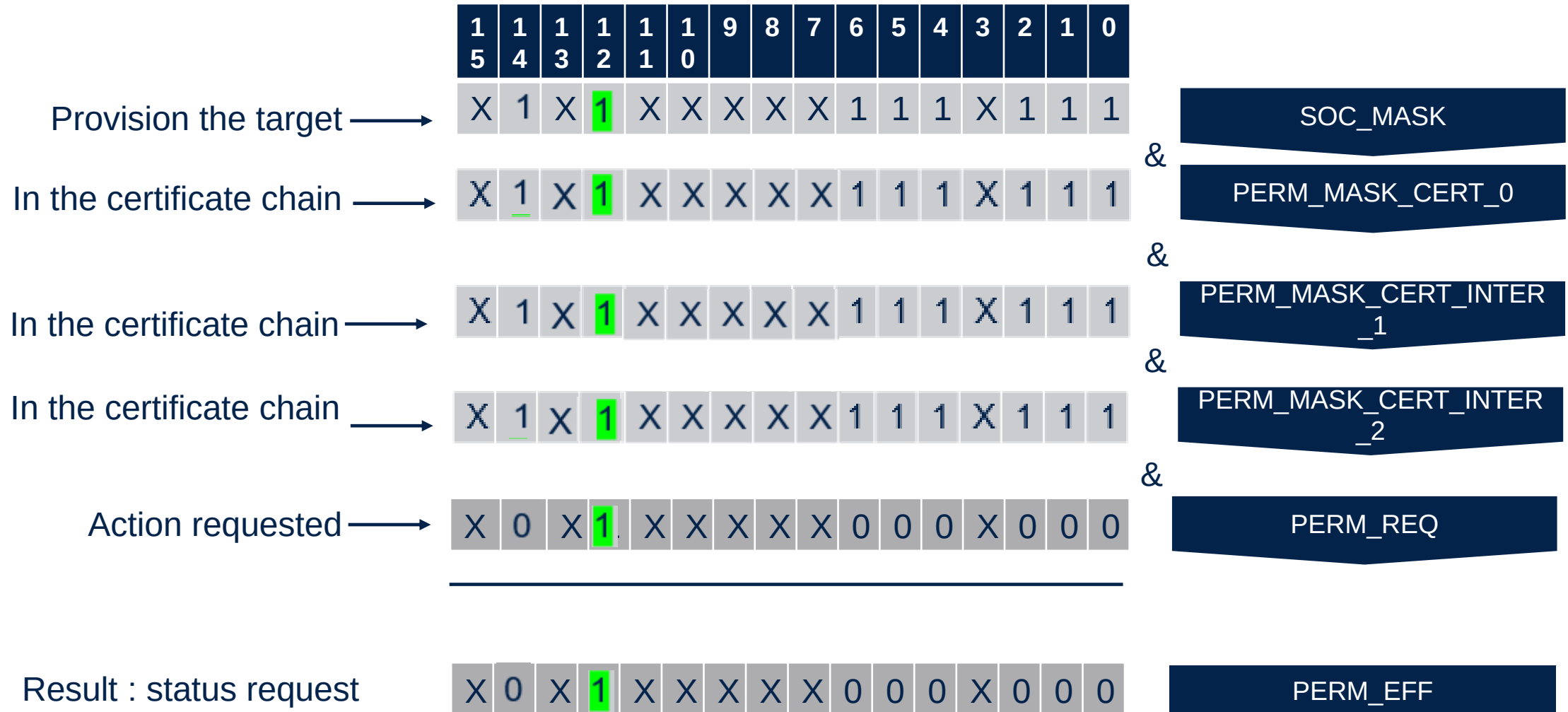
STM32H5 Security Debug Authentication In details





Distribution models

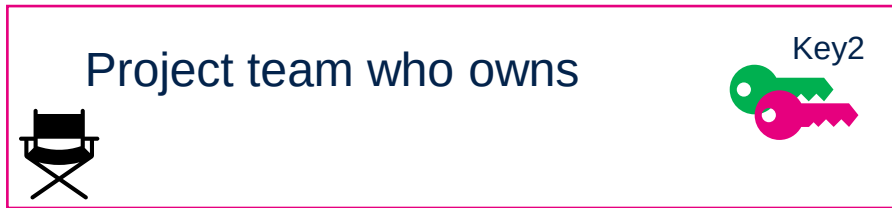
Permissions



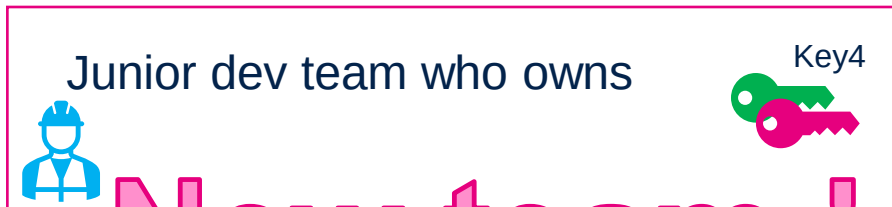


STM32H5 Security

A new team is created...



Role : decide what is the debug authentication capabilities for each team



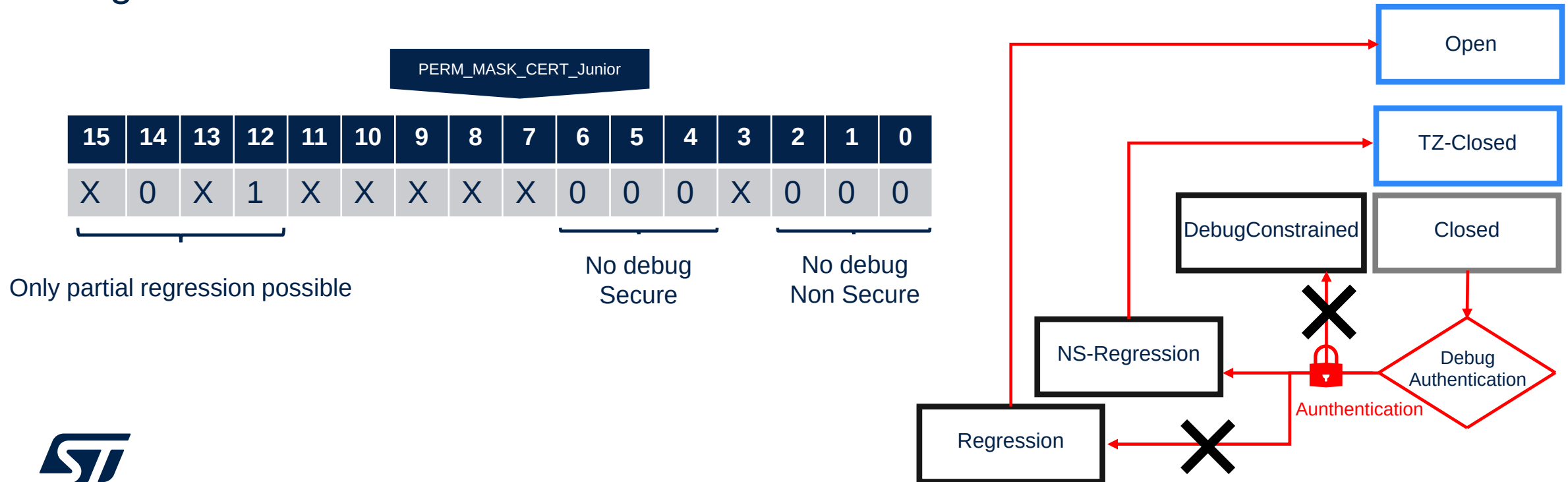
Role : developer of non secure application

New team !



Let's create a new certificate for Junior dev team..

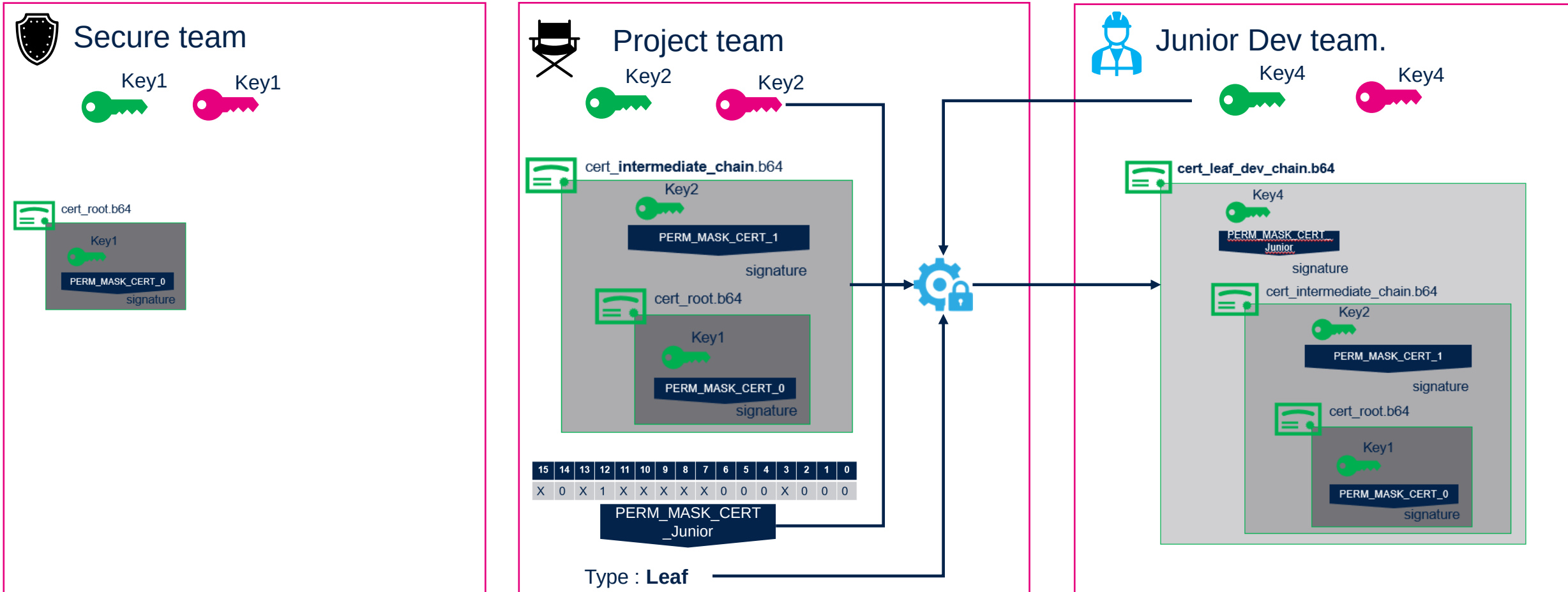
- Project management team doesn't want junior dev team to remove the secure manager from the platform and doesn't allow debug in closed state.
- Project management team create a certificate which allows only the non-secure regression.



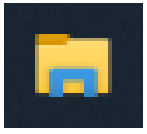


STM32H5 Security

Create a leaf certificate chain



Create a certificate chain for the junior dev team



Copy the junior development keys

Material location : C:\Training\STM32H5_Security_WS\Hands-on-material\TimeStamped_Event_Detection\ROT_Provisioning\DA\Keys

OSDisk (C:) > Training > STM32H5_Security_WS > Hands-on-material > TimeStamped_Event_Detection > ROT_Provisioning > DA > Keys

Name

key_4_leaf.pem
key_4_leaf_pub.pem

COPY

(C:) > Training > STM32H5_Security_WS > Hands-on > TimeStamped_Event_Detection > ROT_Provisioning > DA > Keys

Name

key_1_root.pem
key_1_root_pub.pem
key_2_intermediate.pem
key_2_intermediate_pub.pem
key_3_leaf.pem
key_3_leaf_pub.pem
key_4_leaf.pem
key_4_leaf_pub.pem

Material destination :
C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Keys



Create new certificate

STM32 Trusted Package Creator

File

Edit

Options

Help

Doc

Toolbar icons: Save, Erase, Copy, Log, Cancel

Navigation tabs: OBkey, Image Gen, License Gen, **DA CertifGen**, SFI, SFIx, HSM, SFI Option

Left sidebar (1): SFI, SSP, WB, **H5**

Form fields:

- Microcontroller (3): Refresh
- Certificate Role (4):
- Issuer Private Key: Open
- Leaf Public Key: Open
- Settings



1

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Keys\key_2_intermediate.pem



Certificate Provisioning

Issuer Private Key C:/Training/STM32H5_Security_WS/Hands-on/TimeStamped_Event_Detection/ROT_Provisioning/DA/Keys/key_2_intermediate.pem

Open

Leaf Public Key

C:/Training/STM32H5_Security_WS/Hands-on/TimeStamped_Event_Detection/ROT_Provisioning/DA/Keys/key_4_leaf_pub.pem

Open

Settings

2

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Keys\key_4_leaf_pub.pem



Description

AP - M33

3

Regression				Non-Intrusive Debug				Secure Debug				Non-Secure Debug			
	Full	>To IROT	To TZ		Reserved	Reserved	Reserved		Level 3	Level 2	Level 1		Level 3	Level 2	Level 1
3	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0



Information	
Microcontroller	STM32H5-2M
Role	LEAF
SOC class	0

1

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Certificates\cert_intermediate_chain.b64

Input certificate for chaining

Open

Certificate file

Browse

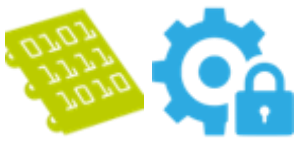
Tool Description:

2

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Certificates\cert_leaf_dev.b64

3

Generate Certificate



Certificate for development team is ready !



OSDisk (C:) > Training > STM32H5_Security_WS > Hands-on > TimeStamped_Event_Detection > ROT_Provisioning > DA > Certificates

Name

cert_leaf_dev.b64
cert_leaf_dev_chain.b64



STM32H5 Security

Junior team Debug Authentication partial regression



Key4

private ECC key



cert_leaf_dev_chain.b64

Key4

PERM_MASK_CERT_Junior

signature

LEAF certificate chain

Request partial regression

Authentication

HASH(Key1_root_pub)
SOC Permission Mask

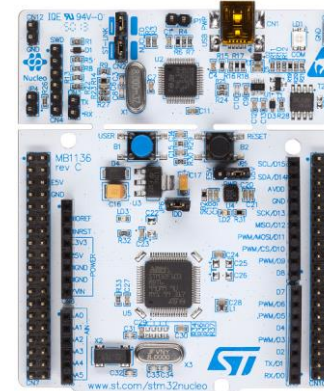


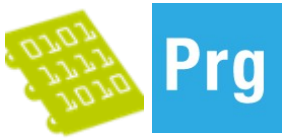
Action requested

Closed



TZ-Closed





Let's first close the debug

If STM32CubeIDE debugger is still connected please disconnect

STM32CubeProgrammer

STM32CubeProgrammer

Data Information Notice

Secure programming

1

2

3

Debug Authentication

This interface supports only STM32H5!

name	value
Locking Mechanism	--
Soc ID	--
Life Cycle	--
Device ID	--

Discover

Close Debug

Used to lock device once the debug has been opened through Debug Authentication process. Applicable only when the feature is available.

Request debug closing

Secure programming

RDP REG

SFI/SFIx

PROV

DA

SSP

Key File Path

Select File

Browse

Certificate File Path

Select File

Browse

Continue

This interface supports only STM32H5!

Discover

name	value
Locking Mechanism	Certificate
Soc ID	0x003A001A 0x3332510F 0x38363236 0x000000...
Life Cycle	PSA_LIFECYCLE_UNKNOWN
Device ID	0x484

1

Close Debug

Used to lock device once the debug has been opened through Debug Authentication process. Applicable only when the feature is available.

Step 1: Path selection.

Step 2: Permission selection

Step 3: Execution.

Message

Target successfully locked.

2

OK

RDP REG
SFI/SFIx
PROV
DA
SSP

Debug Authentication

Key File Path

Select File
Browse

Certificate File Path

Select File
Browse
Continue

This interface supports only STM32H5!

name	value
Locking Mechanism	Certificate
Soc ID	0x003A001A 0x3332510F 0x38363236 0x000000...
Life Cycle	ST_LIFECYCLE_CLOSED
Device ID	0x484

Close Debug

Used to lock device once the debug has been opened through Debug Authentication process. Applicable only when the feature is available.

☒ Step 1: Path selection.
☐ Step 2: Permission selection
☐ Step 3: Execution.



Prg

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Keys\key_4_leaf.pem

RDP RE

Debug Authentication

Key File Path

1 Browse

Certificate File Path

2 Browse

3 Continue

This interface supports only STM32H5!

Discover

name	value
Locking Mechanism	Certificate
Soc ID	0x003A001A 0x3332510F 0x38363236 0x000000...
Life Cycle	ST_LIFECYCLE_CLOSED
Device ID	0x484

Used to lock device once the debug has been authentication process. ... is available.

Step 3: Execution.

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Certificates\cert_leaf_dev_chain.b64

RDP REG
SFI/SFIx
PROV
DA
SSP

Debug Authentication

Key File Path

Select File C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning... Browse

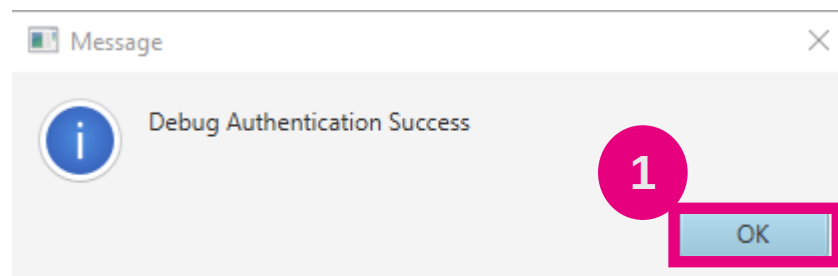
Certificate File Path

Select File C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning... Browse Continue

Permissions

Permission	Select
Non-Secure Intrusive Debug (Level3)	☐
Secure Intrusive Debug (Level1)	☐
Secure Intrusive Debug (Level2)	☐
Secure Intrusive Debug (Level3)	☐
Partial Regression	☒
Full Regression	☐

Execute



Option bytes

3

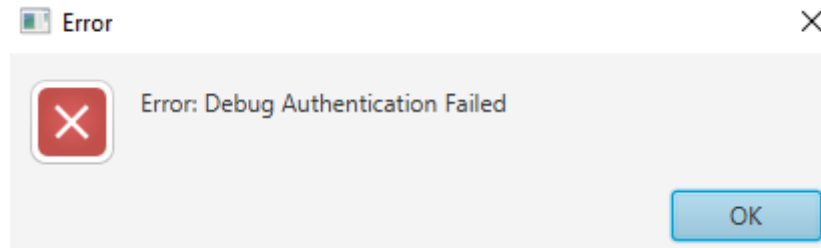
2

Product state

Name	Value	Description
PRODUCT_STATE	C6	Life state code. ED : Open 17 : Provisioning, Debug partially opened (only non-secure) 2E : iRoT-provisioned, Debug partially opened (only non-secure) C6 : TZ-Closed, Debug partially opened (only non-secure) 72 : Closed, Debug disabled, regression is possible 5C : Locked

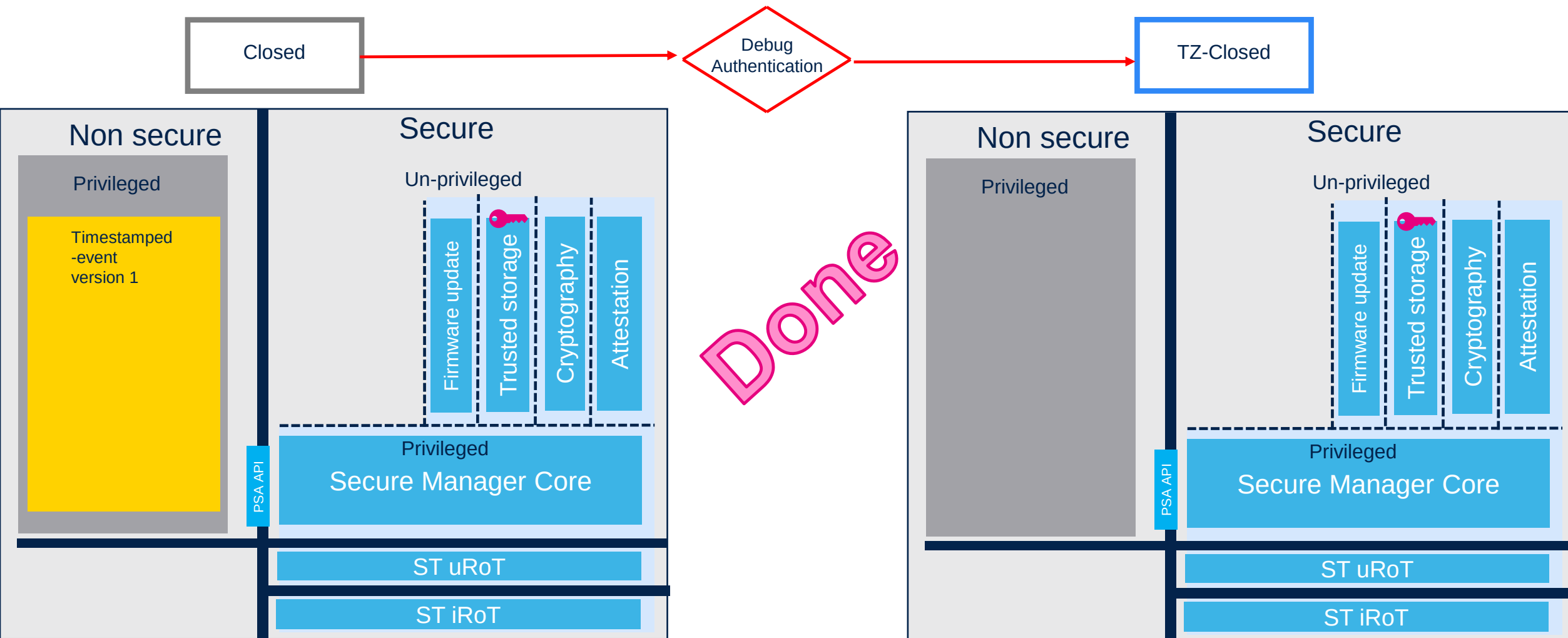


If you try any unauthorized action...



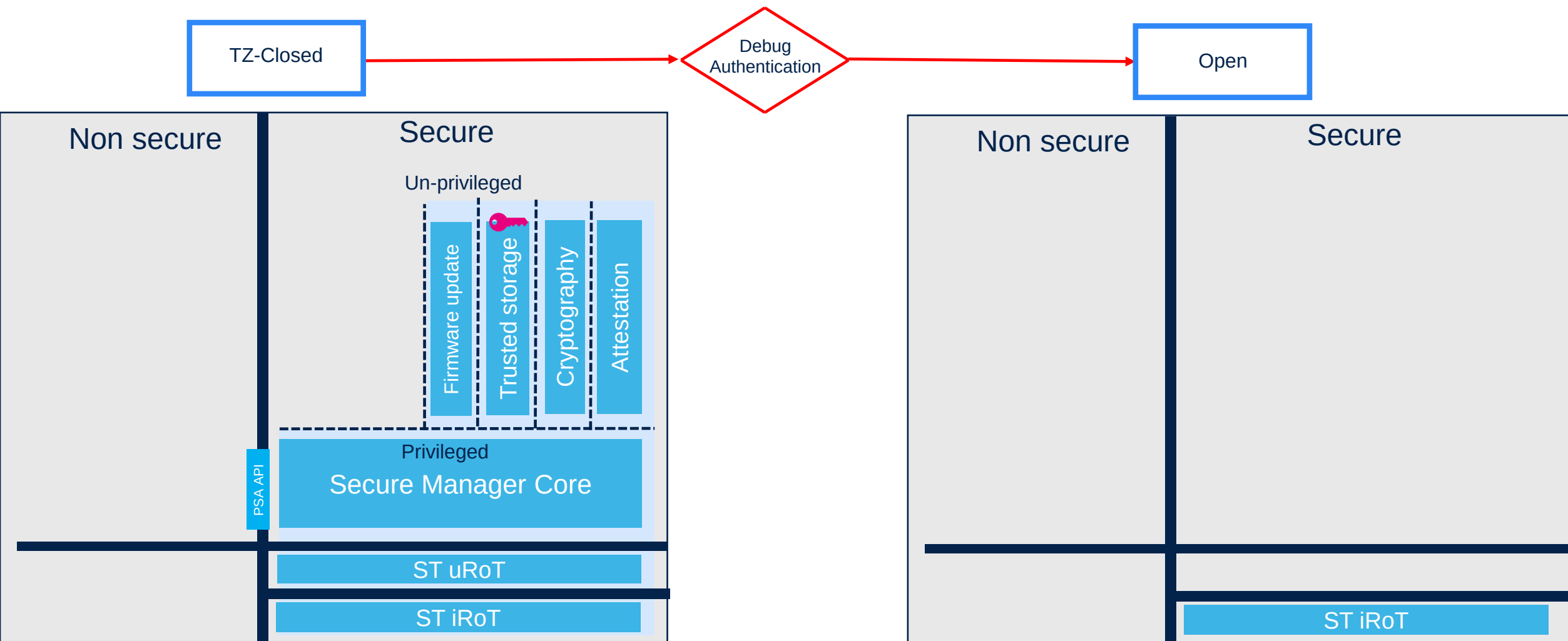


STM32H5 NS-Regression (partial)





STM32H5 Regression



Let's do a full regression

STM32CubeProgrammer

STM32CubeProgrammer

Data Information Notice

Secure programming

1

2

3

Debug Authentication

This interface supports only STM32H5!

Discover

name	value
Locking Mechanism	--
Soc ID	--
Life Cycle	--
Device ID	--

Close Debug

Used to lock device once the debug has been opened through Debug Authentication process. Applicable only when the feature is available.



Prg

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Keys\key_3_leaf.pem

RDP RE

Debug Authentication

Key File Path

1 Browse

Certificate File Path

2 Browse 3 Continue

This interface supports only STM32H5!

Discover

name	value
Locking Mechanism	Certificate
Soc ID	0x003A001A 0x3332510F 0x38363236 0x000000...
Life Cycle	ST_LIFECYCLE_CLOSED
Device ID	0x484

Used to lock device once the debug has been authentication process. Signature is available.

Step 3: Execution.

C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning\DA\Certificates\cert_leaf_chain.b64

RDP REG
SFI/SFIx
PROV
DA
SSP

Debug Authentication

Key File Path

Select File C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning Browse

Certificate File Path

Select File C:\Training\STM32H5_Security_WS\Hands-on\TimeStamped_Event_Detection\ROT_Provisioning Browse Continue

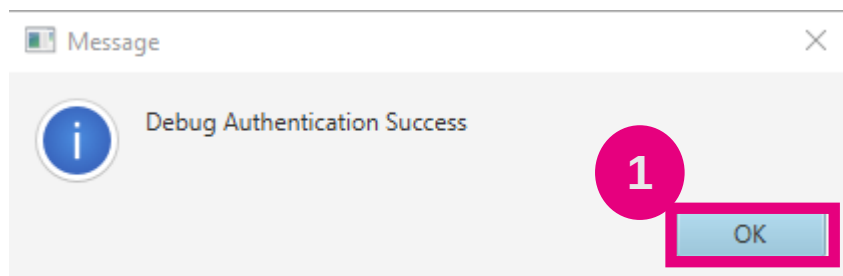
Permissions

Permission	Select
Non-Secure Intrusive Debug (Level3)	☐
Secure Intrusive Debug (Level1)	☐
Secure Intrusive Debug (Level2)	☐
Secure Intrusive Debug (Level3)	☐
Partial Regression	☐
Full Regression	☒

1

2

Execute



3

2

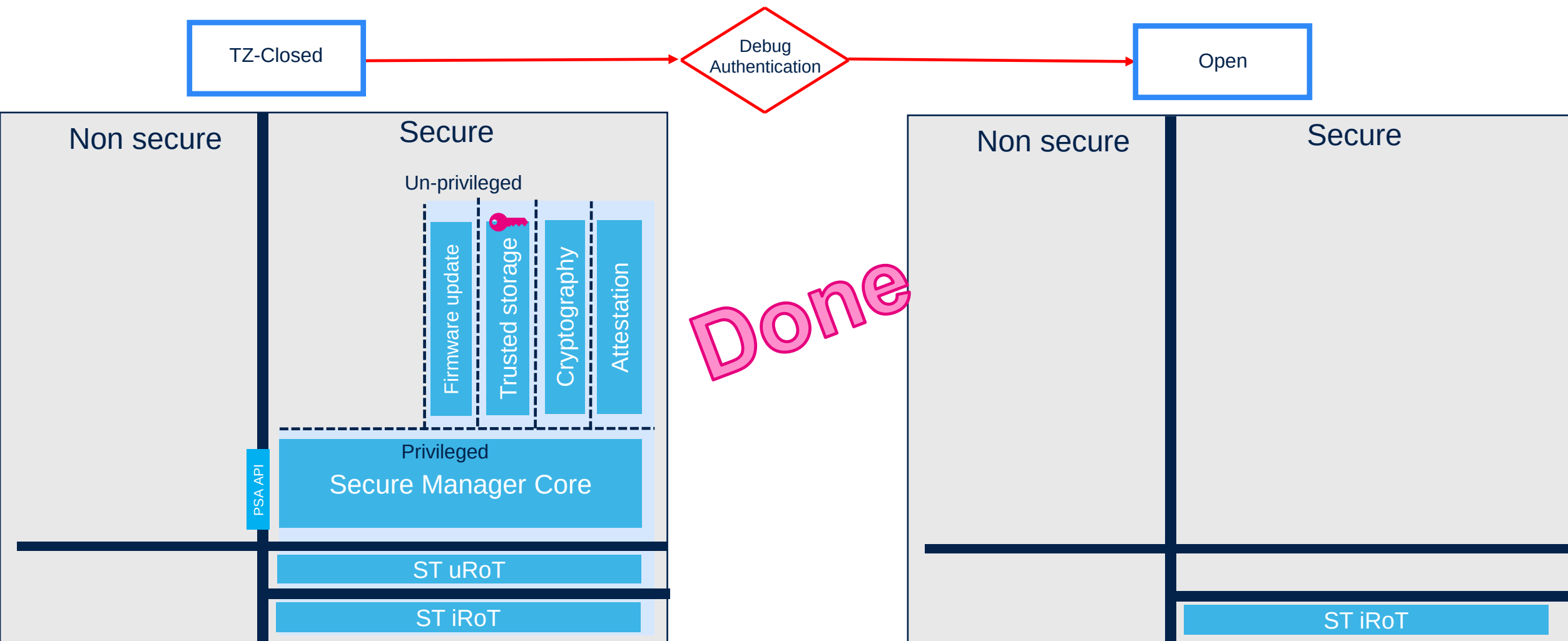
Option bytes

▼ Product state

Name	Value	Description
PRODUCT_STATE	ED ▼	Life state code. ED : Open 17 : Provisioning, Debug partially opened (only non-secure) 2E : iRoT-provisioned, Debug partially opened (only non-secure) C6 : TZ-Closed, Debug partially opened (only non-secure) 72 : Closed, Debug disabled, regression is possible 5C : Locked



STM32H5 Regression





Conclusion

- Debug Authentication is a new feature of STM32H5
- Certificates allow fine control on what user can do with the target
- This feature also simplifies the field return analysis

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